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Virtual transponder utilizing inband commanding

Abstract

Systems, methods, and apparatus for a virtual transponder utilizing inband commanding are disclosed. In one or more embodiments, a disclosed method comprises receiving, by a payload antenna on a vehicle via a hosted receiving antenna, encrypted hosted commands transmitted from a hosted payload (HoP) operation center (HOC). The method further comprises receiving, by the vehicle, encrypted host commands transmitted from a host spacecraft operations center (SOC). Also, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. In addition, the method comprises transmitting, by the payload antenna, payload data to a host receiving antenna and/or the hosted receiving antenna. Additionally, the method comprises transmitting, by a host telemetry transmitter, the encrypted host telemetry to the host SOC. Further, the method comprises transmitting, by a hosted telemetry transmitter, the encrypted hosted telemetry to the HOC via the host SOC.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/227,147, filed Apr. 9, 2021, which is a continuation application of U.S. patent application Ser. No. 16/530,887, filed Aug. 2, 2019, which issued as U.S. Pat. No. 11,388,150 on Jul. 12, 2022, which is a continuation application of U.S. patent application Ser. No. 15/451,267, filed Mar. 6, 2017, which issued as U.S. Pat. No. 10,419,403 on Sep. 17, 2019, the entire disclosures of which are expressly incorporated by reference herein.

FIELD
(1) The present disclosure relates to virtual transponders. In particular, it relates to virtual transponders utilizing inband commanding.
BACKGROUND
(2) Currently, typical transponders on a vehicle (e.g., a satellite) have the ability to perform switching of inputs to outputs of the payload. All of this switching on the payload is commanded and controlled by a single satellite controller with no resource allocation privacy. For example, in a digital transponder, when a user request for a channel with specific bandwidth and antenna characteristics is made, the channel is then set up, used, and then disconnected.
(3) As such, there is a need for an improved transponder design that allows for privacy in the allocation of resources on the payload.
SUMMARY
(4) The present disclosure relates to a method, system, and apparatus for virtual transponders utilizing inband commanding. In one or more embodiments, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method further comprises transmitting, by the hosted receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by a host spacecraft operations center (SOC), encrypted host commands to the vehicle, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Additionally, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. Also, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host

commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to a host receiving antenna and/or the hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. Additionally, the method comprises encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. In addition, the method comprises transmitting, by a host telemetry transmitter, the encrypted host telemetry to the host SOC. Also, the method comprises transmitting, by a hosted telemetry transmitter, the encrypted hosted telemetry to the host SOC. Further, the method comprises transmitting, by the host SOC, the encrypted hosted telemetry to the HOC.

(5) In one or more embodiments, the reconfiguring of the payload according to the unencrypted host commands and/or the unencrypted hosted commands comprises adjusting at least one of: transponder power, transponder spectrum monitoring, transponder connectivity, transponder gain settings, transponder limiter settings, transponder automatic level control settings, transponder phase settings, internal gain generation, bandwidth for at least one beam, at least one frequency band for at least one of the at least one beam, transponder beamforming settings, effective isotropic radiation power (EIRP) for at least one of the at least one beam, transponder channels, and/or beam steering.

(6) In at least one embodiment, the reconfiguring of the payload according to the unencrypted host commands and/or the unencrypted hosted commands comprises reconfiguring at least one of: at least one antenna, at least one analog-to-digital converter, at least one digital-to-analog converter, at least one beamformer, at least one digital channelizer, at least one demodulator, at least one modulator, at least one digital switch matrix, at least one digital combiner, and/or at least one analog switch matrix.

(7) In one or more embodiments, the vehicle is an airborne vehicle. In some embodiments, the airborne vehicle is a satellite, aircraft, unmanned aerial vehicle (UAV), or space plane.

(8) In at least one embodiment, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method further comprises transmitting, by the host SOC, the encrypted hosted commands to a vehicle. Also, the method comprises transmitting, by the host SOC, encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises transmitting, by the host receiving antenna, the encrypted host commands to a payload antenna on the vehicle. Additionally, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or a hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the method comprises encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Additionally, the method comprises transmitting, by a host telemetry transmitter, the encrypted host telemetry to the host SOC. Also, the method comprises transmitting, by a hosted telemetry transmitter, the encrypted hosted telemetry to the host SOC. Further, the method comprises transmitting, by the host SOC, the encrypted hosted telemetry to the HOC.

(9) In one or more embodiments, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method further comprises transmitting, by the hosted receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by a host spacecraft operations center (SOC), encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises transmitting, by the host receiving antenna, the encrypted host commands to a payload antenna on the vehicle. Additionally, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Also, the method comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. In addition, the method comprises encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. Additionally, the method comprises encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Also, the method comprises transmitting, by a host telemetry transmitter, the encrypted host telemetry to the host SOC. In addition, the method comprises transmitting, by a hosted telemetry transmitter, the encrypted hosted telemetry to the host SOC. Further, the method comprises transmitting, by the host SOC, the encrypted hosted telemetry to the HOC.
(10) In at least one embodiment, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method further comprises transmitting, by the hosted receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by a host spacecraft operations center (SOC), encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises transmitting, by the host receiving antenna, the encrypted host commands to the payload antenna on the vehicle. Additionally, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry. In addition, the method comprises transmitting, by a telemetry transmitter, the encrypted telemetry to the host SOC. Further, the method comprises transmitting, by the host SOC, the encrypted telemetry to the HOC.
(11) In one or more embodiments, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method further comprises

transmitting, by the hosted receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by a host spacecraft operations center (SOC), the encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises transmitting, by the host receiving antenna, the encrypted host commands to the payload antenna. Additionally, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the method comprises encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Additionally, the method comprises transmitting, by the payload antenna, the encrypted host telemetry to the host receiving antenna. Also, the method comprises transmitting, by the host receiving antenna, the encrypted host telemetry to the host SOC. In addition, the method comprises transmitting, by the payload antenna, the encrypted hosted telemetry to the hosted receiving antenna. Further, the method comprises transmitting, by the hosted receiving antenna, the encrypted hosted telemetry to the HOC.

(12) In at least one embodiment, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method further comprises transmitting, by the hosted receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by a host spacecraft operations center (SOC), encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises transmitting, by the host receiving antenna, the encrypted host commands to the payload antenna. Additionally, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry. In addition, the method comprises transmitting, by the payload antenna, the encrypted telemetry to the host receiving antenna. Additionally, the method comprises transmitting, by the host receiving antenna, the encrypted telemetry to the host SOC. Also, the method comprises transmitting, by the payload antenna, the encrypted telemetry to the hosted receiving antenna. Further, the method comprises transmitting, by the hosted receiving antenna, the encrypted telemetry to the HOC.

(13) In one or more embodiments, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a host spacecraft operations center (SOC). In one or more embodiments, the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC)

variety. The method further comprises transmitting, by the host SOC, encrypted host commands and the encrypted hosted commands to a host receiving antenna. In one or more embodiments, the encrypted host commands are encrypted utilizing a first COMSEC variety. Also, the method comprises transmitting, by the host receiving antenna, the encrypted host commands and the encrypted hosted commands to a payload antenna on the vehicle. In addition, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Additionally, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. Also, the method comprises reconfiguring a payload on the vehicle according to at least one of the unencrypted host commands or the unencrypted hosted commands. In addition, the method comprises transmitting, by the payload antenna, payload data to at least one of the host receiving antenna or a hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the method comprises encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Also, the method comprises transmitting, by a host telemetry transmitter, the encrypted host telemetry to the host SOC. In addition, the method comprises transmitting, by a hosted telemetry transmitter, the encrypted hosted telemetry to the host SOC. Further, the method comprises transmitting, by the host SOC, the encrypted hosted telemetry to the HOC.

(14) In one or more embodiments, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system further comprises the hosted receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises a host spacecraft operations center (SOC) to transmit encrypted host commands to the vehicle, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Additionally, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. Also, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. In addition, the system comprises the payload antenna to transmit payload data to a host receiving antenna and/or the hosted receiving antenna. Additionally, the system comprises the first communication security module to encrypt unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. Also, the system comprises the second communication security module to encrypt unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. In addition, the system comprises a host telemetry transmitter to transmit the encrypted host telemetry to the host SOC. Additionally, the system comprises a hosted telemetry transmitter to transmit the encrypted hosted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted hosted telemetry to the HOC.

(15) In at least one embodiment, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system further comprises the host SOC to transmit the encrypted hosted commands to a vehicle. Also, the system comprises the host SOC to transmit encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises

the host receiving antenna to transmit the encrypted host commands to a payload antenna on the vehicle. Additionally, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the system comprises the payload antenna to transmit payload data to the host receiving antenna and/or a hosted receiving antenna. Also, the system comprises the first communication security module to encrypt unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the system comprises the second communication security module to encrypt unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Additionally, the system comprises a host telemetry transmitter to transmit the encrypted host telemetry to the host SOC. Also, the system comprises a hosted telemetry transmitter to transmit the encrypted hosted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted hosted telemetry to the HOC.

(16) In one or more embodiments, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system further comprises the hosted receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises a host spacecraft operations center (SOC) to transmit encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises the host receiving antenna to transmit the encrypted host commands to a payload antenna on the vehicle. Additionally, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the system comprises the payload antenna to transmit payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the system comprises the first communication security module to encrypt unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the system comprises the second communication security module to encrypt unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Additionally, the system comprises a host telemetry transmitter to transmit the encrypted host telemetry to the host SOC. Also, the system comprises a hosted telemetry transmitter to transmit the encrypted hosted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted hosted telemetry to the HOC.

(17) In at least one embodiment, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system further comprises the hosted receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises a host spacecraft operations center (SOC) to transmit encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises the host receiving antenna to transmit the encrypted host commands to the payload antenna on the vehicle. Additionally, the system comprises a first communication security module to decrypt the encrypted host commands utilizing

the first COMSEC variety to generate unencrypted host commands. Also, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the system comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the system comprises the first communication security module to encrypt unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry. In addition, the system comprises a telemetry transmitter to transmit the encrypted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted telemetry to the HOC.

(18) In one or more embodiments, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system further comprises the hosted receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises a host spacecraft operations center (SOC) to transmit the encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises the host receiving antenna to transmit the encrypted host commands to the payload antenna. Additionally, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the system comprises the payload antenna to transmit payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the system comprises the first communication security module to encrypt unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the system comprises the second communication security module to encrypt unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Additionally, the system comprises the payload antenna to transmit the encrypted host telemetry to the host receiving antenna. Also, the system comprises the host receiving antenna to transmit the encrypted host telemetry to the host SOC. In addition, the system comprises the payload antenna to transmit the encrypted hosted telemetry to the hosted receiving antenna. Further, the system comprises the hosted receiving antenna to transmit the encrypted hosted telemetry to the HOC.

(19) In at least one embodiment, a system for a virtual transponder utilizing inband commanding comprising a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a hosted receiving antenna, where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system further comprises the hosted receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises a host spacecraft operations center (SOC) to transmit encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises the host receiving antenna to transmit the encrypted host commands to the payload antenna. Additionally, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the system comprises the payload antenna to transmit

payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the system comprises the first communication security module to encrypt unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry. In addition, the system comprises the payload antenna to transmit the encrypted telemetry to the host receiving antenna. Additionally, the system comprises the host receiving antenna to transmit the encrypted telemetry to the host SOC. Also, the system comprises the payload antenna to transmit the encrypted telemetry to the hosted receiving antenna. Further, the system comprises the hosted receiving antenna to transmit the encrypted telemetry to the HOC.

(20) In one or more embodiments, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a host spacecraft operations center (SOC). In one or more embodiments, the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system further comprises the host SOC to transmit encrypted host commands and the encrypted hosted commands to a host receiving antenna. In one or more embodiments, the encrypted host commands are encrypted utilizing a first COMSEC variety. Also, the system comprises the host receiving antenna to transmit the encrypted host commands and the encrypted hosted commands to a payload antenna on the vehicle. In addition, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Additionally, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. Also, the system comprises a payload on the vehicle reconfigured according to at least one of the unencrypted host commands or the unencrypted hosted commands. In addition, the system comprises the payload antenna to transmit payload data to at least one of the host receiving antenna or a hosted receiving antenna. Additionally, the system comprises the first communication security module to encrypt unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the system comprises the second communication security module to encrypt unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Also, the system comprises a host telemetry transmitter to transmit the encrypted host telemetry to the host SOC. In addition, the system comprises a hosted telemetry transmitter to transmit the encrypted hosted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted hosted telemetry to the HOC.

(21) In one or more embodiments, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method also comprises transmitting, by the host SOC, the encrypted hosted commands to a host receiving antenna. The method further comprises transmitting, by the host receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by the host SOC, encrypted host commands to the vehicle, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Additionally, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. Also, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to a host receiving antenna and/or the hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. Additionally,

the method comprises encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. In addition, the method comprises transmitting, by a host telemetry transmitter, the encrypted host telemetry to the host SOC. Also, the method comprises transmitting, by a hosted telemetry transmitter, the encrypted hosted telemetry to the host SOC. Further, the method comprises transmitting, by the host SOC, the encrypted hosted telemetry to the HOC.

(22) In one or more embodiments, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system also comprises the host SOC to transmit the encrypted hosted commands to a host receiving antenna. The system further comprises the host receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises the host SOC to transmit encrypted host commands to the vehicle, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Additionally, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. Also, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. In addition, the system comprises the payload antenna to transmit payload data to a host receiving antenna and/or the hosted receiving antenna. Additionally, the system comprises the first communication security module to encrypt unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. Also, the system comprises the second communication security module to encrypt unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. In addition, the system comprises a host telemetry transmitter to transmit the encrypted host telemetry to the host SOC. Additionally, the system comprises a hosted telemetry transmitter to transmit the encrypted hosted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted hosted telemetry to the HOC.

(23) In at least one embodiment, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method also comprises transmitting, by the host SOC, the encrypted hosted commands to a host receiving antenna. The method further comprises transmitting, by the host receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by the host SOC, encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises transmitting, by the host receiving antenna, the encrypted host commands to the payload antenna on the vehicle. Additionally, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry. In addition, the method comprises transmitting, by a telemetry transmitter, the encrypted telemetry to the host SOC. Further, the method comprises

transmitting, by the host SOC, the encrypted telemetry to the HOC.

(24) In at least one embodiment, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system also comprises the host SOC to transmit the encrypted hosted commands to a host receiving antenna. The system further comprises the host receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises the host SOC to transmit encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises the host receiving antenna to transmit the encrypted host commands to the payload antenna on the vehicle. Additionally, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the system comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the system comprises the first communication security module to encrypt unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry. In addition, the system comprises a telemetry transmitter to transmit the encrypted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted telemetry to the HOC.

(25) In one or more embodiments, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method also comprises transmitting, by the host SOC, the encrypted hosted commands to a host receiving antenna. The method further comprises transmitting, by the host receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by the host SOC, the encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises transmitting, by the host receiving antenna, the encrypted host commands to the payload antenna. Additionally, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the method comprises encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Additionally, the method comprises transmitting, by the payload antenna, the encrypted host telemetry to the host receiving antenna. Also, the method comprises transmitting, by the host receiving antenna, the encrypted host telemetry to the host SOC. In addition, the method comprises transmitting, by the payload antenna, the encrypted hosted telemetry to the host receiving antenna. Also, the method comprises transmitting, by the host receiving antenna, the encrypted hosted telemetry to the host SOC. Further, the method comprises the host SOC, transmitting, the encrypted hosted telemetry to the

HOC.

(26) In one or more embodiments, a system for a virtual transponder utilizing inband commanding comprises a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system also comprises the host SOC to transmit the encrypted hosted commands to a host receiving antenna. The system further comprises the host receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises a host spacecraft operations center (SOC) to transmit the encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises the host receiving antenna to transmit the encrypted host commands to the payload antenna. Additionally, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the system comprises the payload antenna to transmit payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the system comprises the first communication security module to encrypt unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry. In addition, the system comprises the second communication security module to encrypt unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry. Additionally, the system comprises the payload antenna to transmit the encrypted host telemetry to the host receiving antenna. Also, the system comprises the host receiving antenna to transmit the encrypted host telemetry to the host SOC. In addition, the system comprises the payload antenna to transmit the encrypted hosted telemetry to the host receiving antenna. Also, the system comprises the host receiving antenna to transmit the encrypted hosted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted hosted telemetry to the HOC.

(27) In at least one embodiment, a method for a virtual transponder utilizing inband commanding comprises transmitting, by a hosted payload (HoP) operation center (HOC), encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The method also comprises transmitting by the host SOC the encrypted hosted commands to a host receiving antenna. The method further comprises transmitting, by the host receiving antenna, the encrypted hosted commands to a payload antenna on a vehicle. Also, the method comprises transmitting, by the host SOC, encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the method comprises transmitting, by the host receiving antenna, the encrypted host commands to the payload antenna. Additionally, the method comprises decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the method comprises decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the method comprises reconfiguring a payload on the vehicle according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the method comprises transmitting, by the payload antenna, payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the method comprises encrypting, by the first communication security module, unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry. In addition, the method comprises transmitting, by the payload antenna, the encrypted telemetry to the host receiving antenna. Additionally, the method

comprises transmitting, by the host receiving antenna, the encrypted telemetry to the host SOC. Further, the method comprises transmitting, by the host SOC, the encrypted telemetry to the HOC.
(28) In at least one embodiment, a system for a virtual transponder utilizing inband commanding comprising a hosted payload (HoP) operation center (HOC) to transmit encrypted hosted commands to a host spacecraft operations center (SOC), where the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety. The system also comprises the host SOC to transmit the encrypted hosted commands to a host receiving antenna. The system further comprises the host receiving antenna to transmit the encrypted hosted commands to a payload antenna on a vehicle. Also, the system comprises the host SOC to transmit encrypted host commands to a host receiving antenna, where the encrypted host commands are encrypted utilizing a first COMSEC variety. In addition, the system comprises the host receiving antenna to transmit the encrypted host commands to the payload antenna. Additionally, the system comprises a first communication security module to decrypt the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands. Also, the system comprises a second communication security module to decrypt the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands. In addition, the system comprises a payload on the vehicle reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. Additionally, the system comprises the payload antenna to transmit payload data to the host receiving antenna and/or the hosted receiving antenna. Also, the system comprises the first communication security module to encrypt unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry. In addition, the system comprises the payload antenna to transmit the encrypted telemetry to the host receiving antenna. Additionally, the system comprises the host receiving antenna to transmit the encrypted telemetry to the host SOC. Further, the system comprises the host SOC to transmit the encrypted telemetry to the HOC.
(29) In at least one embodiment, a method for a virtual transponder on a vehicle comprises generating, by a configuration algorithm (CA), a configuration for a portion of a payload on the vehicle utilized by a host user by using an option for each of at least one variable for the portion of the payload on the vehicle utilized by the host user. The method further comprises generating, by the CA, a configuration for a portion of the payload on the vehicle utilized by a hosted user by using an option for each of at least one variable for the portion of the payload on the vehicle utilized by the hosted user. Also, the method comprises generating, by a host command generator, host commands for reconfiguring the portion of the payload on the vehicle utilized by the host user by using the configuration for the portion of the payload on the vehicle utilized by the host user. In addition, the method comprises generating, by a hosted command generator, hosted commands for reconfiguring the portion of the payload on the vehicle utilized by the hosted user by using the configuration for the portion of the payload on the vehicle utilized by the hosted user. Additionally, the method comprises transmitting the host commands and the hosted commands to the vehicle. Also, the method comprises reconfiguring the portion of the payload on the vehicle utilized by the host user by using the host commands. Further the method comprises reconfiguring the portion of the payload on the vehicle utilized by the hosted user by using the hosted commands.
(30) In one or more embodiments, at least one variable is: at least one transponder power, at least one transponder spectrum, at least one transponder gain setting, at least one transponder limiter setting, at least one transponder automatic level control setting, at least one transponder phase setting, at least one internal gain generation, bandwidth for at least one beam, at least one frequency band for at least one of at least one beam, at least one transponder beamforming setting, effective isotropic radiation power (EIRP) for at least one of at least one beam, at least one transponder channel, and/or beam steering for at least one of at least one beam.
(31) In at least one embodiment, the reconfiguring comprises reconfiguring: at least one antenna, at least one analog-to-digital converter, at least one digital-to-analog converter, at least one

beamformer, at least one digital channelizer, at least one demodulator, at least one modulator, at least one digital switch matrix, at least one digital combiner, and/or at least one analog switch matrix.
(32) In one or more embodiments, at least one antenna a parabolic reflector antenna, a shaped reflector antenna, a multifeed array antenna, and/or a phased array antenna.
(33) In at least one embodiment, the host computing device and the hosted computing device are located at a respective station. In some embodiments, the station a ground station, a terrestrial vehicle, an airborne vehicle, or a marine vehicle.
(34) In one or more embodiments, the vehicle is an airborne vehicle. In some embodiments, the airborne vehicle is a satellite, an aircraft, an unmanned aerial vehicle (UAV), or a space plane.
(35) In at least one embodiment, the method further comprises selecting, with a host graphical user interface (GUI) on a host computing device, the option for each of at least one variable for the portion of the payload on the vehicle utilized by the host user.
(36) In one or more embodiments, the method further comprises selecting, with a hosted GUI on a hosted computing device, the option for each of at least one variable for the portion of the payload on the vehicle utilized by the hosted user.
(37) In at least one embodiment, a system for a virtual transponder on a vehicle comprises a configuration algorithm (CA) to generate a configuration for a portion of a payload on the vehicle utilized by a host user by using an option for each of at least one variable for the portion of the payload on the vehicle utilized by the host user, and to generate a configuration for a portion of the payload on the vehicle utilized by a hosted user by using an option for each of at least one variable for the portion of the payload on the vehicle utilized by the hosted user. The system further comprises a host command generator to generate host commands for reconfiguring the portion of the payload on the vehicle utilized by the host user by using the configuration for the portion of the payload on the vehicle utilized by the host user. Further, the system comprises a hosted command generator to generate hosted commands for reconfiguring the portion of the payload on the vehicle utilized by the hosted user by using the configuration for the portion of the payload on the vehicle utilized by the hosted user. In one or more embodiments, the portion of the payload on the vehicle utilized by the host user is reconfigured by using the host commands. In some embodiments, the portion of the payload on the vehicle utilized by the hosted user is reconfigured by using the hosted commands.
(38) In one or more embodiments, the system further comprises a host graphical user interface (GUI), on a host computing device, used to select the option for each of at least one variable for the portion of the payload on the vehicle utilized by the host user.
(39) In at least one embodiment, the system further comprises a hosted GUI, on a hosted computing device, used to select the option for each of at least one variable for the portion of the payload on the vehicle utilized by the hosted user.
(40) The features, functions, and advantages can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments.

Description

DRAWINGS
(1) These and other features, aspects, and advantages of the present disclosure will become better understood with regard to the following description, appended claims, and accompanying drawings where:
(2) FIG. **1** is a diagram showing simplified architecture for the disclosed system for a virtual transponder, in accordance with at least one embodiment of the present disclosure.
(3) FIGS. **2**A-**13**H show exemplary systems and methods for a virtual transponder utilizing inband

commanding, in accordance with at least one embodiment of the present disclosure.
(4) FIG. **2**A is a diagram showing the disclosed system for a virtual transponder utilizing inband commanding for the hosted user using a hosted receiving antenna, in accordance with at least one embodiment of the present disclosure.
(5) FIG. **2**B is a diagram showing the disclosed system for a virtual transponder utilizing inband commanding for the hosted user using a host receiving antenna, in accordance with at least one embodiment of the present disclosure.
(6) FIGS. **3**A, **3**B, **3**C, and **3**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the hosted user using a hosted receiving antenna, in accordance with at least one embodiment of the present disclosure.
(7) FIGS. **3**E, **3**F, **3**G, and **3**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the hosted user using a host receiving antenna, in accordance with at least one embodiment of the present disclosure.
(8) FIG. **4** is a diagram showing the disclosed system for a virtual transponder utilizing inband commanding for the host user, in accordance with at least one embodiment of the present disclosure.
(9) FIGS. **5**A, **5**B, **5**C, and **5**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user, in accordance with at least one embodiment of the present disclosure.
(10) FIG. **6**A is a diagram showing the disclosed system for a virtual transponder utilizing inband commanding for the host user and the hosted user using two receiving antennas and employing two communication security (COMSEC) varieties for telemetry, in accordance with at least one embodiment of the present disclosure.
(11) FIG. **6**B is a diagram showing the disclosed system for a virtual transponder utilizing inband commanding for the host user and the hosted user using one receiving antenna and employing two communication security (COMSEC) varieties for telemetry, in accordance with at least one embodiment of the present disclosure.
(12) FIGS. **7**A, **7**B, **7**C, and **7**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user and the hosted user using two receiving antennas and employing two COMSEC varieties for telemetry, in accordance with at least one embodiment of the present disclosure.
(13) FIGS. **7**E, **7**F, **7**G, and **7**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user and the hosted user using one receiving antenna and employing two communication security (COMSEC) varieties for telemetry, in accordance with at least one embodiment of the present disclosure.
(14) FIG. **8**A is a diagram showing the disclosed system for a virtual transponder utilizing inband commanding for the host user and the hosted user using two receiving antennas and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure.
(15) FIG. **8**B is a diagram showing the disclosed system for a virtual transponder utilizing inband commanding for the host user and the hosted user using one receiving antenna and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure.
(16) FIGS. **9**A, **9**B, **9**C, and **9**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user and the hosted user using two receiving antennas and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure.
(17) FIGS. **9**E, **9**F, **9**G, and **9**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user and the hosted user using one receiving antenna and employing one COMSEC variety for telemetry, in accordance with at least one

embodiment of the present disclosure.
(18) FIG. **10**A is a diagram showing the disclosed system for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using two receiving antennas and employing two COMSEC varieties for telemetry, in accordance with at least one embodiment of the present disclosure.
(19) FIG. **10**B is a diagram showing the disclosed system for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using one receiving antenna and employing two COMSEC varieties for telemetry, in accordance with at least one embodiment of the present disclosure.
(20) FIGS. **11**A, **11**B, **11**C, and **11**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using two receiving antennas and employing two COMSEC varieties for telemetry, in accordance with at least one embodiment of the present disclosure.
(21) FIGS. **11**E, **11**F, **11**G, and **11**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using one receiving antenna and employing two COMSEC varieties for telemetry, in accordance with at least one embodiment of the present disclosure.
(22) FIG. **12**A is a diagram showing the disclosed system for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using two receiving antennas and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure.
(23) FIG. **12**B is a diagram showing the disclosed system for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using one receiving antenna and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure.
(24) FIGS. **13**A, **13**B, **13**C, and **13**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using two receiving antennas and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure.
(25) FIGS. **13**E, **13**F, **13**G, and **13**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using one receiving antenna and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure.
(26) FIGS. **14-19**B show exemplary systems and methods for a virtual transponder, in accordance with at least one embodiment of the present disclosure.
(27) FIG. **14** is a diagram showing the disclosed system for a virtual transponder on a vehicle, in accordance with at least one embodiment of the present disclosure.
(28) FIG. **15** is a diagram showing an exemplary allocation of bandwidth amongst a plurality of beams when utilizing the disclosed virtual transponder, in accordance with at least one embodiment of the present disclosure.
(29) FIG. **16** is a diagram showing the switch architecture for a flexible allocation of bandwidth amongst a plurality of beams when utilizing the disclosed virtual transponder, in accordance with at least one embodiment of the present disclosure.
(30) FIG. **17** is a diagram showing details of the digital channelizer of FIG. **16**, in accordance with at least one embodiment of the present disclosure.
(31) FIG. **18** is a diagram showing exemplary components on the vehicle that may be utilized by the disclosed virtual transponder, in accordance with at least one embodiment of the present disclosure.
(32) FIGS. **19**A and **19**B together show a flow chart for the disclosed method for a virtual transponder on a vehicle, in accordance with at least one embodiment of the present disclosure.

DESCRIPTION

(33) The methods and apparatus disclosed herein provide an operative system for virtual transponders utilizing inband commanding. The system of the present disclosure allows for vehicle operators to privately share vehicle resources. It should be noted that in this disclosure, in-band frequency band(s) refer to a frequency band(s) that is the same frequency band(s) utilized to transmit payload data; and out-of-band frequency band(s) refer to a frequency band(s) that is not the same frequency band(s) utilized to transmit payload data.

(34) As previously mentioned above, currently, typical transponders on a vehicle (e.g., a satellite) have the ability to perform switching of inputs to outputs of the payload. All of this switching on the payload is commanded and controlled by a single satellite controller with no resource allocation privacy. For example, in a digital transponder, when a user request for a channel with specific bandwidth and antenna characteristics is made, the channel is then set up, used, and then disconnected.

(35) The disclosed system allows for private vehicle resource allocation and control that provides vehicle users the ability to privately, dynamically, allocate resources on demand. In particular, the disclosed system employs a virtual transponder, which is a transponder partitioned into multiple transponders with independent command and control. In one or more embodiments, an exemplary virtual transponder includes a digital transponder with a digital channelizer, a digital switch matrix, and a digital combiner that is configured to partition a digital transponder into multiple transponders with independent command and control. Command and control of the virtual transponder is achieved via ground software that provides dynamic allocation and privatization of the digital switch matrix for bandwidth on demand.

(36) It should be noted that the disclosed system for private vehicle resource allocation and control may employ various different types of transponders for the virtual transponder other than the specific disclosed embodiments (e.g., depicted FIGS. **16-18**) for the virtual transponder. For example, various different types of transponders may be employed for the virtual transponder including, but not limited to, various different types of digital transponders, various different types of analog transponders (e.g., conventional repeater-type transponders), and various different types of combination analog/digital transponders.

(37) In the following description, numerous details are set forth in order to provide a more thorough description of the system. It will be apparent, however, to one skilled in the art, that the disclosed system may be practiced without these specific details. In the other instances, well known features have not been described in detail so as not to unnecessarily obscure the system.

(38) Embodiments of the present disclosure may be described herein in terms of functional and/or logical components and various processing steps. It should be appreciated that such components may be realized by any number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components (e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like), which may carry out a variety of functions under the control of one or more processors, microprocessors, or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with other components, and that the system described herein is merely one example embodiment of the present disclosure.

(39) For the sake of brevity, conventional techniques and components related to satellite communication systems, and other functional aspects of the system (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

(40) FIG. **1** is a diagram **100** showing simplified architecture for the disclosed system for a virtual transponder, in accordance with at least one embodiment of the present disclosure. In this figure, a simplified view of multiple possible hosted payload configurations is illustrated. In particular, this figure shows a space segment **110** and a ground segment **120**. The space segment **110** represents a vehicle. Various different types of vehicles may be employed for the vehicle including, but not limited to, an airborne vehicle. And, various different types of airborne vehicles may be employed for the vehicle including, but not limited to, a satellite, an aircraft, an unmanned aerial vehicle (UAV), and a space plane.
(41) In the case of a satellite being employed for the vehicle, it should be noted that satellites typically include computer-controlled systems. A satellite generally includes a bus **130** and a payload **140**. The bus **130** may include systems (which include components) that control the satellite. These systems perform tasks, such as power generation and control, thermal control, telemetry, attitude control, orbit control, and other suitable operations.
(42) The payload **140** of the satellite provides functions to users of the satellite. The payload **140** may include antennas, transponders, and other suitable devices. For example, with respect to communications, the payload **140** in a satellite may be used to provide Internet access, telephone communications, radio, television, and other types of communications.
(43) The payload **140** of the satellite may be used by different entities. For example, the payload **140** may be used by the owner of the satellite (i.e. the host user), one or more customers (i.e. the hosted user(s)), or some combination thereof.
(44) For example, the owner of a satellite may lease different portions of the payload **140** to different customers. In one example, one group of antenna beams generated by the payload **140** of the satellite may be leased to one customer, while a second group of antenna beams may be leased to a second customer. In another example, one group of antenna beams generated by the payload **140** of the satellite may be utilized by the owner of the satellite, while a second group of antenna beams may be leased to a customer. In yet another example, some or all of the antenna beams generated by the payload **140** of the satellite may be shared by one customer and a second customer. In another example, some or all of the antenna beams generated by the payload **140** of the satellite may be shared by the owner of the satellite and a customer. When satellites are shared by different users, users may have a shared communications link (e.g., Interface A) to the satellite, or each user may have a separate communications link (e.g., Interfaces A and D) to the satellite.
(45) Leasing a satellite to multiple customers may increase the revenues that an owner of a satellite can obtain. Further, a customer may use a subset of the total resources in a satellite for a cost that is less than the cost for the customer to purchase and operate a satellite, to build and operate a satellite, or to lease an entire satellite.
(46) Referring back to FIG. **1**, the ground segment **120** comprises a host spacecraft operations center (SOC) (e.g., a ground station associated with the owner of the satellite) **150**, and a hosted payload (HoP) operation center(s) (HOC(s)) (e.g., a ground station(s) associated with a customer(s) that is leasing at least a portion of the payload of the satellite from the owner) **160**.
(47) FIG. **1** shows a number of different possible communication links (i.e. Interfaces A-E). It should be noted that the disclosed system may employ some or all of these illustrated communication links. Interface A, which may comprise multiple links, is an out-of-band command and telemetry link from the host SOC **150** to command the satellite. Interface B, which may comprise multiple links, is a communication link, between the bus **130** and the payload **140**. Interface B may be used to control essential items, such as power. Information that may be communicated from the bus **130** to the payload **140** via Interface B may include, but is not limited to, time, ephemeris, and payload commands. Information that may be communicated from the payload **140** to the bus **130** via Interface B may include, but is not limited to, payload telemetry.
(48) Interface C, which may comprise multiple links, is an inband command and telemetry link for bus and/or payload. Interface D, which may comprise multiple links, is a command and telemetry

link from the HOC(s) **160** to command the satellite. Interface E, which may comprise multiple links, between the host SOC **150** and the HOCs **160** allows for requests from the HOCs for resource sharing of the payload **140**.

(49) FIGS. **2**A-**13**H show exemplary systems and methods for a virtual transponder utilizing inband commanding, in accordance with at least one embodiment of the present disclosure.

(50) FIG. **2**A is a diagram **200** showing the disclosed system for a virtual transponder utilizing inband commanding for the hosted user (i.e. HOC) **260** using a hosted receiving antenna **290**, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **210**, a host SOC **250**, and a HOC **260** are shown. The HOC **260** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **205** of the vehicle **210** from the owner of a satellite (i.e. the host SOC) **250**. It should be noted that in some embodiments, the HOC **260** may lease all of the payload **205** of the vehicle **210** from the owner of a satellite (i.e. the host SOC) **250**. Also, it should be noted that is some embodiments, the HOC **260** may own the payload **205** (e.g., a steerable antenna) of the vehicle **210**, and contract the host SOC **250** to transmit encrypted hosted commands to the vehicle **210**.

(51) During operation, the HOC **260** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **205** that the HOC **260** is leasing from the host SOC **250**. The host SOC **250** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **205** that host SOC **250** is utilizing for itself.

(52) It should be noted that, although in FIG. **2**A the host SOC **250** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **250** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).

(53) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).

(54) The HOC **260** then transmits **215** the encrypted hosted commands to a hosted receiving antenna **290**. After the hosted receiving antenna **290** receives the encrypted hosted commands, the hosted receiving antenna **290** transmits **297** the encrypted hosted commands to a payload antenna **280** on the vehicle **210**. The hosted receiving antenna **290** transmits **297** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **280** transmits the encrypted hosted commands to a payload **205**. The payload **205** on the vehicle **210** receives the encrypted hosted commands. The payload **205** then transmits **255** the encrypted hosted commands to a second communication security module **265**. The second communication security module **265** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.

(55) It should be noted that the second communication security module **265** may comprise one or more modules. In addition, the second communication security module **265** may comprise one or more processors.

(56) The host SOC **250** transmits **220** the encrypted host commands to the vehicle **210**. The host SOC **250** transmits **220** the encrypted host commands utilizing an out-of-band frequency band(s) (i.e. a frequency band(s) that is not the same frequency band(s) utilized to transmit payload data). The host command receiver **235** on the vehicle **210** receives the encrypted host commands.

(57) The host command receiver **235** then transmits **252** the encrypted host commands to a first communication security module **262**. The first communication security module **262** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.
(58) It should be noted that the first communication security module **262** may comprise one or more modules. In addition, the first communication security module **262** may comprise one or more processors.
(59) The first communication security module **262** then transmits **270** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **205**. The second communication security module **265** transmits **275** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **205**. The payload **205** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **280** then transmits (e.g., in one or more antenna beams **281**) payload data to a host receiving antenna **285** and/or the hosted receiving antenna **290** on the ground.
(60) Also, it should be noted that, although in FIG. **2**A, antenna beams **281** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **281** may include more or less number of beams than is shown in FIG. **2**A (e.g., antenna beams **281** may only include a single beam), and antenna beams **281** may include beams of different shapes than circular spot beams as is shown in FIG. **2**A (e.g., antenna beams **281** may include elliptical beams and/or shaped beams of various different shapes).
(61) It should be noted that in one or more embodiments, the payload antenna **280** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **280** may comprise one or more multifeed antenna arrays.
(62) The payload **205** transmits **291** unencrypted host telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **205** that is utilized by the host SOC **250**) to the first communication security module **262**. The first communication security module **262** then encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry (i.e. encrypted host TLM).
(63) The payload **205** transmits **292** unencrypted hosted telemetry (i.e. unencrypted HoP TLM, which is telemetry data related to the portion of the payload **205** that is leased by the HOC **260**) to the second communication security module **265**. The second communication security module **265** then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry (i.e. encrypted HoP TLM).
(64) The first communication security module **262** then transmits **293** the encrypted host telemetry to a host telemetry transmitter **289**. The host telemetry transmitter **289** then transmits **295** the encrypted host telemetry to the host SOC **250**. The host telemetry transmitter **289** transmits **295** the encrypted host telemetry utilizing an out-of-band frequency band(s). The host SOC **250** then decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry.
(65) The second communication security module **265** then transmits **296** the encrypted hosted telemetry to a hosted telemetry transmitter **294**. The hosted telemetry transmitter **294** then transmits **298** the encrypted hosted telemetry to the host SOC **250**. The telemetry transmitter **294** transmits **298** the encrypted hosted telemetry utilizing an out-of-band frequency band(s). The host SOC **250** then transmits **299** the encrypted hosted telemetry to the HOC **260**. The HOC **260** then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry.
(66) FIG. **2**B is a diagram **2000** showing the disclosed system for a virtual transponder utilizing inband commanding for the hosted user (i.e. HOC) **2060** using a host receiving antenna **20085**, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **2010**, a

host SOC **2050**, and a HOC **2060** are shown. The HOC **2060** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **2005** of the vehicle **2010** from the owner of a satellite (i.e. the host SOC) **2050**. It should be noted that in some embodiments, the HOC **2060** may lease all of the payload **2005** of the vehicle **2010** from the owner of a satellite (i.e. the host SOC) **2050**. Also, it should be noted that is some embodiments, the HOC **2060** may own the payload **2005** (e.g., a steerable antenna) of the vehicle **2010**, and contract the host SOC **2050** to transmit encrypted hosted commands to the vehicle **2010**.

(67) During operation, the HOC **2060** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **2005** that the HOC **2060** is leasing from the host SOC **2050**. The host SOC **2050** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **2005** that host SOC **2050** is utilizing for itself.

(68) It should be noted that, although in FIG. 2B the host SOC **2050** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **2050** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).

(69) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).

(70) The HOC **2060** then transmits **2015** the encrypted hosted commands to the host SOC **2050**. Then, the host SOC **2050** transmits **2016** the encrypted hosted commands to a host receiving antenna **2085**. After the host receiving antenna **2085** receives the encrypted hosted commands, the host receiving antenna **2085** transmits **2097** the encrypted hosted commands to a payload antenna **2080** on the vehicle **2010**. The host receiving antenna **2085** transmits **2097** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **2080** transmits the encrypted hosted commands to a payload **2005**. The payload **2005** on the vehicle **2010** receives the encrypted hosted commands. The payload **2005** then transmits **2055** the encrypted hosted commands to a second communication security module **2065**. The second communication security module **2065** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.

(71) It should be noted that the second communication security module **2065** may comprise one or more modules. In addition, the second communication security module **2065** may comprise one or more processors.

(72) The host SOC **2050** transmits **2020** the encrypted host commands to the vehicle **2010**. The host SOC **2050** transmits **2020** the encrypted host commands utilizing an out-of-band frequency band(s) (i.e. a frequency band(s) that is not the same frequency band(s) utilized to transmit payload data). The host command receiver **2035** on the vehicle **2010** receives the encrypted host commands.

(73) The host command receiver **2035** then transmits **2052** the encrypted host commands to a first communication security module **2062**. The first communication security module **2062** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.

(74) It should be noted that the first communication security module **2062** may comprise one or more modules. In addition, the first communication security module **2062** may comprise one or more processors.

(75) The first communication security module **2062** then transmits **2070** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **2005**. The second communication security module **2065** transmits **2075** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **2005**. The payload **2005** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **2080** then transmits (e.g., in one or more antenna beams **2081**) payload data to a host receiving antenna **2085** and/or the hosted receiving antenna **2090** on the ground.
(76) Also, it should be noted that, although in FIG. **2**B, antenna beams **2081** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **2081** may include more or less number of beams than is shown in FIG. **2**B (e.g., antenna beams **2081** may only include a single beam), and antenna beams **2081** may include beams of different shapes than circular spot beams as is shown in FIG. **2**B (e.g., antenna beams **2081** may include elliptical beams and/or shaped beams of various different shapes).
(77) It should be noted that in one or more embodiments, the payload antenna **2080** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **2080** may comprise one or more multifeed antenna arrays.
(78) The payload **2005** transmits **2091** unencrypted host telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **2005** that is utilized by the host SOC **2050**) to the first communication security module **2062**. The first communication security module **2062** then encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry (i.e. encrypted host TLM).
(79) The payload **2005** transmits **2092** unencrypted hosted telemetry (i.e. unencrypted HoP TLM, which is telemetry data related to the portion of the payload **2005** that is leased by the HOC **2060**) to the second communication security module **2065**. The second communication security module **2065** then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry (i.e. encrypted HoP TLM).
(80) The first communication security module **2062** then transmits **2093** the encrypted host telemetry to a host telemetry transmitter **2089**. The host telemetry transmitter **2089** then transmits **2095** the encrypted host telemetry to the host SOC **2050**. The host telemetry transmitter **2089** transmits **2095** the encrypted host telemetry utilizing an out-of-band frequency band(s). The host SOC **2050** then decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry.
(81) The second communication security module **2065** then transmits **2096** the encrypted hosted telemetry to a hosted telemetry transmitter **2094**. The hosted telemetry transmitter **2094** then transmits **2098** the encrypted hosted telemetry to the host SOC **2050**. The telemetry transmitter **2094** transmits **2098** the encrypted hosted telemetry utilizing an out-of-band frequency band(s). The host SOC **2050** then transmits **2099** the encrypted hosted telemetry to the HOC **2060**. The HOC **2060** then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry.
(82) FIGS. **3**A, **3**B, **3**C, and **3**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the hosted user using a hosted receiving antenna, in accordance with at least one embodiment of the present disclosure. At the start **300** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **305**. Then, the HOC transmits the encrypted hosted commands to a hosted receiving antenna **310**. The hosted receiving antenna then transmits the encrypted hosted commands to a payload antenna on a vehicle **315**. Then, the payload antenna transmits the encrypted hosted commands to a payload **320**. The payload then transmits the encrypted hosted commands to a second communication security module **325**.

(83) Then, a host spacecraft operations center (SOC) encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **330**. The host SOC then transmits the encrypted host commands to the vehicle **335**. Then, a host command receiver on the vehicle receives the encrypted host commands **340**. The host command receiver then transmits the encrypted host commands to a first communication security module **345**. Then, the first communication security module decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **350**. The second communication security module decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **355**. Then, the first communication security module transmits the unencrypted host commands to the payload **360**. The second communication security module transmits the unencrypted hosted commands to the payload **365**.

(84) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **370**. The payload antenna then transmits payload data to a host receiving antenna and/or the hosted receiving antenna **375**. Then, the payload transmits to the first communication security module unencrypted host telemetry **380**. The payload transmits to the second communication security module unencrypted hosted telemetry **385**. Then, the first communication security module encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry **390**. The second communication security module encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry **391**.

(85) Then, the first communication security module transmits the encrypted host telemetry to a host telemetry transmitter **392**. The host telemetry transmitter then transmits the encrypted host telemetry to the host SOC **393**. Then, the host SOC decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **394**.

(86) Then, the second communication security module transmits the encrypted hosted telemetry to a hosted telemetry transmitter **395**. The hosted telemetry transmitter then transmits the encrypted hosted telemetry to the host SOC **396**. Then, the host SOC transmits the encrypted hosted telemetry to the HOC **397**. The HOC then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry **398**. Then, the method ends **399**.

(87) FIGS. **3**E, **3**F, **3**G, and **3**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the hosted user using a host receiving antenna, in accordance with at least one embodiment of the present disclosure. At the start **3000** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **3005**. Then, the HOC transmits the encrypted hosted commands to a host spacecraft operations center (SOC) **3010**. The host SOC then transmits the encrypted hosted commands to a host receiving antenna **3012**. The host receiving antenna then transmits the encrypted hosted commands to a payload antenna on a vehicle **3015**. Then, the payload antenna transmits the encrypted hosted commands to a payload **3020**. The payload then transmits the encrypted hosted commands to a second communication security module **3025**.

(88) Then, a host SOC encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **3030**. The host SOC then transmits the encrypted host commands to the vehicle **3035**. Then, a host command receiver on the vehicle receives the encrypted host commands **3040**. The host command receiver then transmits the encrypted host commands to a first communication security module **3045**. Then, the first communication security module decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **3050**. The second communication security module decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **3055**. Then, the first communication security module transmits the unencrypted host commands to the payload **3060**. The second communication security module transmits the

unencrypted hosted commands to the payload **3065**.
(89) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **3070**. The payload antenna then transmits payload data to a host receiving antenna and/or the hosted receiving antenna **3075**. Then, the payload transmits to the first communication security module unencrypted host telemetry **3080**. The payload transmits to the second communication security module unencrypted hosted telemetry **3085**. Then, the first communication security module encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry **3090**. The second communication security module encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry **3091**.
(90) Then, the first communication security module transmits the encrypted host telemetry to a host telemetry transmitter **3092**. The host telemetry transmitter then transmits the encrypted host telemetry to the host SOC **3093**. Then, the host SOC decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **3094**.
(91) Then, the second communication security module transmits the encrypted hosted telemetry to a hosted telemetry transmitter **3095**. The hosted telemetry transmitter then transmits the encrypted hosted telemetry to the host SOC **3096**. Then, the host SOC transmits the encrypted hosted telemetry to the HOC **3097**. The HOC then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry **3098**. Then, the method ends **3099**.
(92) FIG. **4** is a diagram **400** showing the disclosed system for a virtual transponder utilizing inband commanding for the host user (i.e. host SOC) **450**, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **410**, a host SOC **450**, and a HOC **460** are shown. The HOC **460** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **405** of the vehicle **410** from the owner of a satellite (i.e. the host SOC) **450**. It should be noted that in some embodiments, the HOC **460** may lease all of the payload **405** of the vehicle **410** from the owner of a satellite (i.e. the host SOC) **450**. Also, it should be noted that is some embodiments, the HOC **460** may own the payload **405** (e.g., a steerable antenna) of the vehicle **410**, and contract the host SOC **450** to transmit encrypted hosted commands to the vehicle **410**.
(93) During operation, the HOC **460** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **405** that the HOC **460** is leasing from the host SOC **450**. The host SOC **450** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **405** that host SOC **450** is utilizing for itself.
(94) It should be noted that, although in FIG. **4** the host SOC **450** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **450** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).
(95) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).
(96) The host SOC **450** then transmits **420** the encrypted host commands to a host receiving antenna **485**. After the host receiving antenna **485** receives the encrypted host commands, the host receiving antenna **485** transmits **497** the encrypted host commands to a payload antenna **480** on the vehicle **410**. The host receiving antenna **485** transmits **497** the encrypted host commands utilizing

an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **480** transmits the encrypted host commands to a payload **405**. The payload **405** on the vehicle **410** receives the encrypted host commands. The payload **405** then transmits **452** the encrypted host commands to a first communication security module **462**. The first communication security module **462** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.
(97) It should be noted that the first communication security module **462** may comprise one or more modules. In addition, the first communication security module **462** may comprise one or more processors.
(98) The HOC **460** transmits **415** the encrypted hosted commands to the host SOC **450**. The host SOC **450** transmits **425** the encrypted hosted commands to the vehicle **410**. The host SOC **450** transmits **425** the encrypted hosted commands utilizing an out-of-band frequency band(s) (i.e. a frequency band(s) that is not the same frequency band(s) utilized to transmit payload data). The hosted command receiver **435** on the vehicle **410** receives the encrypted host commands.
(99) The hosted command receiver **435** then transmits **455** the encrypted hosted commands to a second communication security module **465**. The second communication security module **465** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.
(100) It should be noted that the second communication security module **465** may comprise one or more modules. In addition, the second communication security module **465** may comprise one or more processors.
(101) The first communication security module **462** then transmits **470** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **405**. The second communication security module **465** transmits **475** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **405**. The payload **405** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **480** then transmits (e.g., in one or more antenna beams **481**) payload data to a host receiving antenna **485** and/or the hosted receiving antenna **490** on the ground.
(102) Also, it should be noted that, although in FIG. **4**, antenna beams **481** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **481** may include more or less number of beams than is shown in FIG. **4** (e.g., antenna beams **481** may only include a single beam), and antenna beams **481** may include beams of different shapes than circular spot beams as is shown in FIG. **4** (e.g., antenna beams **481** may include elliptical beams and/or shaped beams of various different shapes).
(103) It should be noted that in one or more embodiments, the payload antenna **480** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **480** may comprise one or more multifeed antenna arrays.
(104) The payload **405** transmits **491** unencrypted host telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **405** that is utilized by the host SOC **450**) to the first communication security module **462**. The first communication security module **462** then encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry (i.e. encrypted host TLM).
(105) The payload **405** transmits **492** unencrypted hosted telemetry (i.e. unencrypted HoP TLM, which is telemetry data related to the portion of the payload **405** that is leased by the HOC **460**) to the second communication security module **465**. The second communication security module **465** then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry (i.e. encrypted HoP TLM).
(106) The first communication security module **462** then transmits **493** the encrypted host

telemetry to a host telemetry transmitter **489**. The host telemetry transmitter **489** then transmits **495** the encrypted host telemetry to the host SOC **450**. The host telemetry transmitter **489** transmits **495** the encrypted host telemetry utilizing an out-of-band frequency band(s). The host SOC **450** then decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry.

(107) The second communication security module **465** then transmits **496** the encrypted hosted telemetry to a hosted telemetry transmitter **494**. The hosted telemetry transmitter **494** then transmits **498** the encrypted hosted telemetry to the host SOC **450**. The telemetry transmitter **494** transmits **498** the encrypted hosted telemetry utilizing an out-of-band frequency band(s). The host SOC **450** then transmits **499** the encrypted hosted telemetry to the HOC **460**. The HOC **460** then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry.

(108) FIGS. **5**A, **5**B, **5**C, and **5**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user, in accordance with at least one embodiment of the present disclosure. At the start **500** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **505**. Then, the HOC transmits the encrypted hosted commands to a host spacecraft operations center (SOC) **510**. The host SOC then transmits the encrypted hosted commands to a vehicle **515**. Then, a hosted command receiver on the vehicle receives the encrypted hosted commands **520**. The hosted command receiver then transmits the encrypted hosted commands to a second communication security module **525**.

(109) Then, the host SOC encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **530**. The host SOC then transmits the encrypted host commands to a host receiving antenna **535**. Then, the host receiving antenna transmits the encrypted host commands to a payload antenna on the vehicle **540**. The payload antenna then transmits the encrypted host commands to a payload on the vehicle **545**. Then, the payload transmits the encrypted host commands to a first communication security module **550**. The first communication security module then decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **555**. The second communication security module decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **560**. Then, the first communication security module transmits the unencrypted host commands to the payload **565**. The second communication security module transmits the unencrypted hosted commands to the payload **570**.

(110) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **575**. The payload antenna then transmits payload data to the host receiving antenna and/or a hosted receiving antenna **580**. Then, the payload transmits to the first communication security module unencrypted host telemetry **585**. The payload transmits to the second communication security module unencrypted hosted telemetry **590**. Then, the first communication security module encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry **591**. The second communication security module encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry **592**.

(111) Then, the first communication security module transmits the encrypted host telemetry to a host telemetry transmitter **593**. The host telemetry transmitter then transmits the encrypted host telemetry to the host SOC **594**. Then, the host SOC decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **595**.

(112) Then, the second communication security module transmits the encrypted hosted telemetry to a hosted telemetry transmitter **596**. The hosted telemetry transmitter then transmits the encrypted hosted telemetry to the host SOC **597**. Then, the host SOC transmits the encrypted hosted telemetry

to the HOC **598**. The HOC then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry **599**. Then, the method ends **501**.

(113) FIG. **6**A is a diagram **600** showing the disclosed system for a virtual transponder utilizing inband commanding for the host user (i.e. host SOC) **650** and the hosted user (i.e. HOC) **660** using two receiving antennas and employing two communication security (COMSEC) varieties for telemetry, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **610**, a host SOC **650**, and a HOC **660** are shown. The HOC **660** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **605** of the vehicle **610** from the owner of a satellite (i.e. the host SOC) **650**. It should be noted that in some embodiments, the HOC **660** may lease all of the payload **605** of the vehicle **610** from the owner of a satellite (i.e. the host SOC) **650**. Also, it should be noted that is some embodiments, the HOC **660** may own the payload **605** (e.g., a steerable antenna) of the vehicle **610**, and contract the host SOC **650** to transmit encrypted hosted commands to the vehicle **610**.

(114) During operation, the HOC **660** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **605** that the HOC **660** is leasing from the host SOC **650**. The host SOC **650** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **605** that host SOC **650** is utilizing for itself.

(115) It should be noted that, although in FIG. **6**A the host SOC **650** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **650** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).

(116) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).

(117) The host SOC **650** then transmits **620** the encrypted host commands to a host receiving antenna **685**. After the host receiving antenna **685** receives the encrypted host commands, the host receiving antenna **685** transmits **621** the encrypted host commands to a payload antenna **680** on the vehicle **610**. The host receiving antenna **685** transmits **621** the encrypted host commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **680** transmits the encrypted host commands to a payload **605**. The payload **605** on the vehicle **610** receives the encrypted host commands. The payload **605** then transmits **652** the encrypted host commands to a first communication security module **662**. The first communication security module **662** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.

(118) It should be noted that the first communication security module **662** may comprise one or more modules. In addition, the first communication security module **662** may comprise one or more processors.

(119) The HOC **660** then transmits **615** the encrypted hosted commands to a hosted receiving antenna **690**. After the hosted receiving antenna **690** receives the encrypted hosted commands, the hosted receiving antenna **690** transmits **697** the encrypted hosted commands to a payload antenna **680** on the vehicle **610**. The hosted receiving antenna **690** transmits **697** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **680** transmits the

encrypted hosted commands to a payload **605**. The payload **605** on the vehicle **610** receives the encrypted hosted commands. The payload **605** then transmits **655** the encrypted hosted commands to a second communication security module **665**. The second communication security module **665** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.
(120) It should be noted that the second communication security module **665** may comprise one or more modules. In addition, the second communication security module **665** may comprise one or more processors.
(121) The first communication security module **662** then transmits **670** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **605**. The second communication security module **665** transmits **675** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **605**. The payload **605** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **680** then transmits (e.g., in one or more antenna beams **681**) payload data to the host receiving antenna **685** and/or the hosted receiving antenna **690** on the ground.
(122) Also, it should be noted that, although in FIG. **6**A, antenna beams **681** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **681** may include more or less number of beams than is shown in FIG. **6**A (e.g., antenna beams **681** may only include a single beam), and antenna beams **681** may include beams of different shapes than circular spot beams as is shown in FIG. **6**A (e.g., antenna beams **681** may include elliptical beams and/or shaped beams of various different shapes).
(123) It should be noted that in one or more embodiments, the payload antenna **680** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **680** may comprise one or more multifeed antenna arrays.
(124) The payload **605** transmits **691** unencrypted host telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **605** that is utilized by the host SOC **650**) to the first communication security module **662**. The first communication security module **662** then encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry (i.e. encrypted host TLM).
(125) The payload **605** transmits **692** unencrypted hosted telemetry (i.e. unencrypted HoP TLM, which is telemetry data related to the portion of the payload **605** that is leased by the HOC **660**) to the second communication security module **665**. The second communication security module **665** then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry (i.e. encrypted HoP TLM).
(126) The first communication security module **662** then transmits **693** the encrypted host telemetry to a host telemetry transmitter **689**. The host telemetry transmitter **689** then transmits **695** the encrypted host telemetry to the host SOC **650**. The host telemetry transmitter **689** transmits **695** the encrypted host telemetry utilizing an out-of-band frequency band(s). The host SOC **650** then decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry.
(127) The second communication security module **665** then transmits **696** the encrypted hosted telemetry to a hosted telemetry transmitter **694**. The hosted telemetry transmitter **694** then transmits **698** the encrypted hosted telemetry to the host SOC **650**. The telemetry transmitter **694** transmits **698** the encrypted hosted telemetry utilizing an out-of-band frequency band(s). The host SOC **650** then transmits **699** the encrypted hosted telemetry to the HOC **660**. The HOC **660** then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry.
(128) FIG. **6**B is a diagram **6000** showing the disclosed system for a virtual transponder utilizing inband commanding for the host user (i.e. host SOC) **6050** and the hosted user (i.e. HOC) **6060**

using one receiving antenna and employing two communication security (COMSEC) varieties for telemetry, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **6010**, a host SOC **6050**, and a HOC **6060** are shown. The HOC **6060** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **6005** of the vehicle **6010** from the owner of a satellite (i.e. the host SOC) **6050**. It should be noted that in some embodiments, the HOC **6060** may lease all of the payload **6005** of the vehicle **6010** from the owner of a satellite (i.e. the host SOC) **6050**. Also, it should be noted that is some embodiments, the HOC **6060** may own the payload **6005** (e.g., a steerable antenna) of the vehicle **6010**, and contract the host SOC **6050** to transmit encrypted hosted commands to the vehicle **6010**.

(129) During operation, the HOC **6060** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **6005** that the HOC **6060** is leasing from the host SOC **6050**. The host SOC **6050** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **6005** that host SOC **6050** is utilizing for itself.

(130) It should be noted that, although in FIG. **6**B the host SOC **6050** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **6050** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).

(131) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).

(132) The HOC **6060** transmits **6015** the encrypted hosted commands to the host SOC **6050**. The host SOC **6050** then transmits **6020**, **6025** the encrypted host commands and the encrypted hosted commands to a host receiving antenna **6085**. After the host receiving antenna **6085** receives the encrypted host commands and the encrypted hosted commands, the host receiving antenna **6085** transmits **6097**, **6021** the encrypted host commands and the encrypted hosted commands to a payload antenna **6080** on the vehicle **6010**. The host receiving antenna **6085** transmits **6097**, **6021** the encrypted host commands and the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **6080** transmits the encrypted host commands and the encrypted hosted commands to a payload **6005**. The payload **6005** on the vehicle **6010** receives the encrypted host commands and the encrypted hosted commands. The payload **6005** then transmits **6052** the encrypted host commands to a first communication security module **6062**. The first communication security module **6062** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.

(133) It should be noted that the first communication security module **6062** may comprise one or more modules. In addition, the first communication security module **6062** may comprise one or more processors.

(134) The payload **6005** then transmits **6055** the encrypted hosted commands to a second communication security module **6065**. The second communication security module **6065** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.

(135) It should be noted that the second communication security module **6065** may comprise one or more modules. In addition, the second communication security module **6065** may comprise one or more processors.

(136) The first communication security module **6062** then transmits **6070** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **6005**. The second communication security module **6065** transmits **6075** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **6005**. The payload **6005** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **6080** then transmits (e.g., in one or more antenna beams **6081**) payload data to a host receiving antenna **6085** and/or the hosted receiving antenna **6090** on the ground.
(137) Also, it should be noted that, although in FIG. **6**B, antenna beams **6081** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **6081** may include more or less number of beams than is shown in FIG. **6**B (e.g., antenna beams **6081** may only include a single beam), and antenna beams **6081** may include beams of different shapes than circular spot beams as is shown in FIG. **6**B (e.g., antenna beams **6081** may include elliptical beams and/or shaped beams of various different shapes).
(138) It should be noted that in one or more embodiments, the payload antenna **6080** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **6080** may comprise one or more multifeed antenna arrays.
(139) The payload **6005** transmits **6091** unencrypted host telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **6005** that is utilized by the host SOC **6050**) to the first communication security module **6062**. The first communication security module **6062** then encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry (i.e. encrypted host TLM).
(140) The payload **6005** transmits **6092** unencrypted hosted telemetry (i.e. unencrypted HoP TLM, which is telemetry data related to the portion of the payload **6005** that is leased by the HOC **6060**) to the second communication security module **6065**. The second communication security module **6065** then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry (i.e. encrypted HoP TLM).
(141) The first communication security module **6062** then transmits **6093** the encrypted host telemetry to a host telemetry transmitter **6089**. The host telemetry transmitter **6089** then transmits **6095** the encrypted host telemetry to the host SOC **6050**. The host telemetry transmitter **6089** transmits **6095** the encrypted host telemetry utilizing an out-of-band frequency band(s). The host SOC **6050** then decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry.
(142) The second communication security module **6065** then transmits **6096** the encrypted hosted telemetry to a hosted telemetry transmitter **6094**. The hosted telemetry transmitter **6094** then transmits **6098** the encrypted hosted telemetry to the host SOC **6050**. The telemetry transmitter **6094** transmits **6098** the encrypted hosted telemetry utilizing an out-of-band frequency band(s). The host SOC **6050** then transmits **6099** the encrypted hosted telemetry to the HOC **6060**. The HOC **6060** then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry.
(143) FIGS. **7**A, **7**B, **7**C, and **7**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user and the hosted user using two receiving antennas and employing two COMSEC varieties for telemetry, in accordance with at least one embodiment of the present disclosure. At the start **700** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **705**. Then, the HOC transmits the encrypted hosted commands to a hosted receiving antenna **710**. The hosted receiving antenna then transmits the encrypted hosted commands to a payload antenna on a vehicle **715**. Then, the payload antenna transmits the encrypted hosted commands to a payload on the vehicle **720**. The payload then transmits the encrypted hosted commands to a second

communication security module **725**.

(144) Then, a host spacecraft operations center (SOC) encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **730**. The host SOC then transmits the encrypted host commands to a host receiving antenna **735**. Then, the host receiving antenna transmits the encrypted host commands to a payload antenna on the vehicle **740**. The payload antenna then transmits the encrypted host commands to the payload **745**. Then, the payload transmits the encrypted host commands to a first communication security module **750**. The first communication security module then decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **755**. The second communication security module decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **760**. Then, the first communication security module transmits the unencrypted host commands to the payload **765**. The second communication security module then transmits the unencrypted hosted commands to the payload **770**.

(145) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **775**. The payload antenna then transmits payload data to the host receiving antenna and/or the hosted receiving antenna **780**. Then, the payload transmits to the first communication security module unencrypted host telemetry **785**. The payload then transmits to the second communication security module unencrypted hosted telemetry **790**. Then, the first communication security module encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry **791**. The second communication security module encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry **792**.

(146) Then, the first communication security module transmits the encrypted host telemetry to a host telemetry transmitter **793**. The host telemetry transmitter then transmits the encrypted host telemetry to the host SOC **794**. Then, the host SOC decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **795**.

(147) Then, the second communication security module transmits the encrypted hosted telemetry to a hosted telemetry transmitter **796**. The hosted telemetry transmitter then transmits the encrypted hosted telemetry to the host SOC **797**. Then, the host SOC transmits the encrypted hosted telemetry to the HOC **798**. The HOC then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry **799**. Then, the method ends **701**.

(148) FIGS. **7**E, **7**F, **7**G, and **7**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user and the hosted user using one receiving antenna and employing two communication security (COMSEC) varieties for telemetry, in accordance with at least one embodiment of the present disclosure. At the start **70000** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **70005**. Then, the HOC transmits the encrypted hosted commands to a host spacecraft operations center (SOC) **70010**. The host SOC encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **70015**. The host SOC then transmits the encrypted host commands and the encrypted hosted commands to a host receiving antenna **70020**. Then, the host receiving antenna transmits the encrypted host commands and the encrypted hosted commands to a payload antenna on the vehicle **70025**. The payload antenna then transmits the encrypted host commands and the encrypted hosted commands to a payload on the vehicle **70030**. Then, the payload transmits the encrypted host commands to a first communication security module **70035**. The payload then transmits the encrypted hosted commands to a second communication security module **70040**.

(149) The first communication security module then decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **70045**. The second communication security module decrypts the encrypted hosted commands utilizing the second

COMSEC variety to generate the unencrypted hosted commands **70050**. Then, the first communication security module transmits the unencrypted host commands to the payload **70055**. The second communication security module transmits the unencrypted hosted commands to the payload **70060**.

(150) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **70065**. The payload antenna then transmits payload data to the host receiving antenna and/or a hosted receiving antenna **70070**. Then, the payload transmits to the first communication security module unencrypted host telemetry **70075**. The payload transmits to the second communication security module unencrypted hosted telemetry **70080**. Then, the first communication security module encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry **70085**. The second communication security module encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry **70090**.

(151) Then, the first communication security module transmits the encrypted host telemetry to a host telemetry transmitter **70091**. The host telemetry transmitter then transmits the encrypted host telemetry to the host SOC **70092**. Then, the host SOC decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **70093**.

(152) Then, the second communication security module transmits the encrypted hosted telemetry to a hosted telemetry transmitter **70094**. The hosted telemetry transmitter then transmits the encrypted hosted telemetry to the host SOC **70095**. Then, the host SOC transmits the encrypted hosted telemetry to the HOC **70096**. The HOC then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry **70097**. Then, the method ends **70098**.

(153) FIG. **8**A is a diagram **800** showing the disclosed system for a virtual transponder utilizing inband commanding for the host user (i.e. host SOC) **850** and the hosted user (i.e. HOC) **860** using two receiving antennas and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **810**, a host SOC **850**, and a HOC **860** are shown. The HOC **860** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **805** of the vehicle **810** from the owner of a satellite (i.e. the host SOC) **850**. It should be noted that in some embodiments, the HOC **860** may lease all of the payload **805** of the vehicle **810** from the owner of a satellite (i.e. the host SOC) **850**. Also, it should be noted that is some embodiments, the HOC **860** may own the payload **805** (e.g., a steerable antenna) of the vehicle **810**, and contract the host SOC **850** to transmit encrypted hosted commands to the vehicle **810**.

(154) During operation, the HOC **860** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **805** that the HOC **860** is leasing from the host SOC **850**. The host SOC **850** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **805** that host SOC **850** is utilizing for itself.

(155) It should be noted that, although in FIG. **8**A the host SOC **850** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **850** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).

(156) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).

(157) The host SOC **850** then transmits **820** the encrypted host commands to a host receiving antenna **885**. After the host receiving antenna **885** receives the encrypted host commands, the host receiving antenna **885** transmits **821** the encrypted host commands to a payload antenna **880** on the vehicle **810**. The host receiving antenna **885** transmits **821** the encrypted host commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **880** transmits the encrypted host commands to a payload **805**. The payload **805** on the vehicle **810** receives the encrypted host commands. The payload **805** then transmits **852** the encrypted host commands to a first communication security module **862**. The first communication security module **862** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.

(158) It should be noted that the first communication security module **862** may comprise one or more modules. In addition, the first communication security module **862** may comprise one or more processors.

(159) The HOC **860** then transmits **815** the encrypted hosted commands to a hosted receiving antenna **890**. After the hosted receiving antenna **890** receives the encrypted hosted commands, the hosted receiving antenna **890** transmits **897** the encrypted hosted commands to a payload antenna **880** on the vehicle **810**. The hosted receiving antenna **890** transmits **897** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **880** transmits the encrypted hosted commands to a payload **805**. The payload **805** on the vehicle **810** receives the encrypted hosted commands. The payload **805** then transmits **855** the encrypted hosted commands to a second communication security module **865**. The second communication security module **865** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.

(160) It should be noted that the second communication security module **865** may comprise one or more modules. In addition, the second communication security module **865** may comprise one or more processors.

(161) The first communication security module **862** then transmits **870** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **805**. The second communication security module **865** transmits **875** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **805**. The payload **805** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **880** then transmits (e.g., in one or more antenna beams **881**) payload data to the host receiving antenna **885** and/or the hosted receiving antenna **890** on the ground.

(162) Also, it should be noted that, although in FIG. **8**A, antenna beams **881** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **881** may include more or less number of beams than is shown in FIG. **8**A (e.g., antenna beams **881** may only include a single beam), and antenna beams **881** may include beams of different shapes than circular spot beams as is shown in FIG. **8**A (e.g., antenna beams **881** may include elliptical beams and/or shaped beams of various different shapes).

(163) It should be noted that in one or more embodiments, the payload antenna **880** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **880** may comprise one or more multifeed antenna arrays.

(164) The payload **805** transmits **891** unencrypted telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **805** that is utilized by the host SOC **850**; and unencrypted HoP TLM, which is telemetry data related to the portion of the payload **805** that is leased by the HOC **860**) to the first communication security module **862**. The first communication security module **862** then encrypts the unencrypted telemetry utilizing the first COMSEC variety to

generate encrypted telemetry (i.e. encrypted TLM).
(165) The first communication security module **862** then transmits **893** the encrypted telemetry to a telemetry transmitter **889**. The telemetry transmitter **889** then transmits **895** the encrypted telemetry to the host SOC **850**. The host telemetry transmitter **889** transmits **895** the encrypted telemetry utilizing an out-of-band frequency band(s).
(166) The host SOC **850** then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry. The host SOC **850** then utilizes a database that comprises host payload decommutated information and does not comprise hosted payload decommutated information (i.e. a database without hosted payload decommutated information) to read to unencrypted telemetry to determine the telemetry data related to the portion of the payload **805** that is utilized by the host SOC **850**.
(167) The host SOC **850** then transmits **899** the encrypted telemetry to the HOC **860**. The HOC **860** then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry. The HOC **860** then utilizes a database that comprises hosted payload decommutated information and does not comprise host payload decommutated information (i.e. a database without host payload decommutated information) to read to unencrypted telemetry to determine the telemetry data related to the portion of the payload **805** that is utilized by the HOC **860**.
(168) FIG. **8**B is a diagram **8000** showing the disclosed system for a virtual transponder utilizing inband commanding for the host user (i.e. host SOC) **8050** and the hosted user (i.e. HOC) **8060** using one receiving antenna and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **8010**, a host SOC **8050**, and a HOC **8060** are shown. The HOC **8060** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **8005** of the vehicle **8010** from the owner of a satellite (i.e. the host SOC) **8050**. It should be noted that in some embodiments, the HOC **8060** may lease all of the payload **8005** of the vehicle **8010** from the owner of a satellite (i.e. the host SOC) **8050**. Also, it should be noted that is some embodiments, the HOC **8060** may own the payload **8005** (e.g., a steerable antenna) of the vehicle **8010**, and contract the host SOC **8050** to transmit encrypted hosted commands to the vehicle **8010**.
(169) During operation, the HOC **8060** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **8005** that the HOC **8060** is leasing from the host SOC **8050**. The host SOC **8050** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **8005** that host SOC **8050** is utilizing for itself.
(170) It should be noted that, although in FIG. **8**B the host SOC **8050** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **8050** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).
(171) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).
(172) The host SOC **8050** then transmits **8020** the encrypted host commands to a host receiving antenna **8085**. After the host receiving antenna **8085** receives the encrypted host commands, the host receiving antenna **8085** transmits **8021** the encrypted host commands to a payload antenna **8080** on the vehicle **8010**. The host receiving antenna **8085** transmits **8021** the encrypted host

commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **8080** transmits the encrypted host commands to a payload **8005**. The payload **8005** on the vehicle **8010** receives the encrypted host commands. The payload **8005** then transmits **8052** the encrypted host commands to a first communication security module **8062**. The first communication security module **8062** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.

(173) It should be noted that the first communication security module **8062** may comprise one or more modules. In addition, the first communication security module **8062** may comprise one or more processors.

(174) The HOC **8060** then transmits **8015** the encrypted hosted commands to the host SOC **8050**. The host SOC **8050** then transmits **8016** the encrypted hosted commands to a host receiving antenna **8085**. After the host receiving antenna **8085** receives the encrypted hosted commands, the host receiving antenna **8085** transmits **8097** the encrypted hosted commands to a payload antenna **8080** on the vehicle **8010**. The host receiving antenna **8085** transmits **8097** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **8080** transmits the encrypted hosted commands to a payload **8005**. The payload **8005** on the vehicle **8010** receives the encrypted hosted commands. The payload **8005** then transmits **8055** the encrypted hosted commands to a second communication security module **8065**. The second communication security module **8065** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.

(175) It should be noted that the second communication security module **8065** may comprise one or more modules. In addition, the second communication security module **8065** may comprise one or more processors.

(176) The first communication security module **8062** then transmits **8070** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **8005**. The second communication security module **8065** transmits **8075** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **8005**. The payload **8005** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **8080** then transmits (e.g., in one or more antenna beams **8081**) payload data to the host receiving antenna **8085** and/or the hosted receiving antenna **8090** on the ground.

(177) Also, it should be noted that, although in FIG. **8**B, antenna beams **8081** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **8081** may include more or less number of beams than is shown in FIG. **8**B (e.g., antenna beams **8081** may only include a single beam), and antenna beams **8081** may include beams of different shapes than circular spot beams as is shown in FIG. **8**B (e.g., antenna beams **8081** may include elliptical beams and/or shaped beams of various different shapes).

(178) It should be noted that in one or more embodiments, the payload antenna **8080** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **8080** may comprise one or more multifeed antenna arrays.

(179) The payload **8005** transmits **8091** unencrypted telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **8005** that is utilized by the host SOC **8050**; and unencrypted HoP TLM, which is telemetry data related to the portion of the payload **8005** that is leased by the HOC **8060**) to the first communication security module **8062**. The first communication security module **8062** then encrypts the unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry (i.e. encrypted TLM).

(180) The first communication security module **8062** then transmits **8093** the encrypted telemetry to a telemetry transmitter **8089**. The telemetry transmitter **8089** then transmits **8095** the encrypted

telemetry to the host SOC **8050**. The host telemetry transmitter **8089** transmits **8095** the encrypted telemetry utilizing an out-of-band frequency band(s).

(181) The host SOC **8050** then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry. The host SOC **8050** then utilizes a database that comprises host payload decommutated information and does not comprise hosted payload decommutated information (i.e. a database without hosted payload decommutated information) to read to unencrypted telemetry to determine the telemetry data related to the portion of the payload **8005** that is utilized by the host SOC **8050**.

(182) The host SOC **8050** then transmits **8099** the encrypted telemetry to the HOC **8060**. The HOC **8060** then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry. The HOC **8060** then utilizes a database that comprises hosted payload decommutated information and does not comprise host payload decommutated information (i.e. a database without host payload decommutated information) to read to unencrypted telemetry to determine the telemetry data related to the portion of the payload **8005** that is utilized by the HOC **8060**.

(183) FIGS. **9**A, **9**B, **9**C, and **9**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user and the hosted user using two receiving antennas and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure. At the start **900** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **905**. Then, the HOC transmits the encrypted hosted commands to a hosted receiving antenna **910**. The hosted receiving antenna then transmits the encrypted hosted commands to a payload antenna on a vehicle **915**. Then, the payload antenna transmits the encrypted hosted commands to a payload on the vehicle **920**. The payload then transmits the encrypted hosted commands to a second communication security module **925**.

(184) Then, a host spacecraft operations center (SOC) encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **930**. The host SOC then transmits the encrypted host commands to a host receiving antenna **935**. Then, the host receiving antenna transmits the encrypted host commands to a payload antenna on the vehicle **940**. The payload antenna then transmits the encrypted host commands to the payload **945**. Then, the payload transmits the encrypted host commands to a first communication security module **950**. The first communication security module then decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **955**. Then, the second communication security module decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **960**. The first communication security module then transmits the unencrypted host commands to the payload **965**. Then, the second communication security module transmits the unencrypted hosted commands to the payload **970**.

(185) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **975**. The payload antenna then transmits payload data to the host receiving antenna and/or the hosted receiving antenna **980**. Then, the payload transmits to the first communication security module unencrypted telemetry **985**. The first communication security module then encrypts the unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry **990**.

(186) Then, the first communication security module transmits the encrypted telemetry to a telemetry transmitter **991**. The telemetry transmitter then transmits the encrypted telemetry to the host SOC **992**. Then, the host SOC decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **993**. The host SOC then determining telemetry data related to a portion of the payload utilized by the host SOC by using a database without hosted

decommutated information to read the unencrypted telemetry **994**. Then, the host SOC transmits the encrypted telemetry to the HOC **995**. The HOC then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry **996**. Then, the HOC determines telemetry data related to a portion of the payload utilized by the HOC by using a database without host decommutated information to read the unencrypted telemetry **997**. Then, the method ends **998**.
(187) FIGS. **9**E, **9**F, **9**G, and **9**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband commanding for the host user and the hosted user using one receiving antenna and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure. At the start **9000** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **9005**. Then, the HOC transmits the encrypted hosted commands to a host spacecraft operations center (SOC) **9010**. The host SOC then transmits the encrypted hosted commands to a host receiving antenna **9012**. The host receiving antenna then transmits the encrypted hosted commands to a payload antenna on a vehicle **9015**. Then, the payload antenna transmits the encrypted hosted commands to a payload on the vehicle **9020**. The payload then transmits the encrypted hosted commands to a second communication security module **9025**.
(188) Then, a host spacecraft operations center (SOC) encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **9030**. The host SOC then transmits the encrypted host commands to a host receiving antenna **9035**. Then, the host receiving antenna transmits the encrypted host commands to a payload antenna on the vehicle **9040**. The payload antenna then transmits the encrypted host commands to the payload **9045**. Then, the payload transmits the encrypted host commands to a first communication security module **9050**. The first communication security module then decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **9055**. Then, the second communication security module decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **9060**. The first communication security module then transmits the unencrypted host commands to the payload **9065**. Then, the second communication security module transmits the unencrypted hosted commands to the payload **9070**.
(189) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **9075**. The payload antenna then transmits payload data to the host receiving antenna and/or the hosted receiving antenna **9080**. Then, the payload transmits to the first communication security module unencrypted telemetry **9085**. The first communication security module then encrypts the unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry **9090**.
(190) Then, the first communication security module transmits the encrypted telemetry to a telemetry transmitter **9091**. The telemetry transmitter then transmits the encrypted telemetry to the host SOC **9092**. Then, the host SOC decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **9093**. The host SOC then determining telemetry data related to a portion of the payload utilized by the host SOC by using a database without hosted decommutated information to read the unencrypted telemetry **9094**. Then, the host SOC transmits the encrypted telemetry to the HOC **9095**. The HOC then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry **9096**. Then, the HOC determines telemetry data related to a portion of the payload utilized by the HOC by using a database without host decommutated information to read the unencrypted telemetry **9097**. Then, the method ends **9098**.
(191) FIG. **10**A is a diagram **1000** showing the disclosed system for a virtual transponder utilizing inband telemetry and commanding for the host user (i.e. host SOC) **1050** and the hosted user (i.e. HOC) **1060** using two receiving antennas and employing two COMSEC varieties for telemetry, in

accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **1010**, a host SOC **1050**, and a HOC **1060** are shown. The HOC **1060** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **1005** of the vehicle **1010** from the owner of a satellite (i.e. the host SOC) **1050**. It should be noted that in some embodiments, the HOC **1060** may lease all of the payload **1005** of the vehicle **1010** from the owner of a satellite (i.e. the host SOC) **1050**. Also, it should be noted that is some embodiments, the HOC **1060** may own the payload **1005** (e.g., a steerable antenna) of the vehicle **1010**, and contract the host SOC **1050** to transmit encrypted hosted commands to the vehicle **1010**.

(192) During operation, the HOC **1060** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **1005** that the HOC **1060** is leasing from the host SOC **1050**. The host SOC **1050** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **1005** that host SOC **1050** is utilizing for itself.

(193) It should be noted that, although in FIG. **10**A the host SOC **1050** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **1050** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).

(194) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).

(195) The host SOC **1050** then transmits **1020** the encrypted host commands to a host receiving antenna **1085**. After the host receiving antenna **1085** receives the encrypted host commands, the host receiving antenna **1085** transmits **1021** the encrypted host commands to a payload antenna **1080** on the vehicle **1010**. The host receiving antenna **1085** transmits **1021** the encrypted host commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **1080** transmits the encrypted host commands to a payload **1005**. The payload **1005** on the vehicle **1010** receives the encrypted host commands. The payload **1005** then transmits **1052** the encrypted host commands to a first communication security module **1062**. The first communication security module **1062** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.

(196) It should be noted that the first communication security module **1062** may comprise one or more modules. In addition, the first communication security module **1062** may comprise one or more processors.

(197) The HOC **1060** then transmits **1015** the encrypted hosted commands to a hosted receiving antenna **1090**. After the hosted receiving antenna **1090** receives the encrypted hosted commands, the hosted receiving antenna **1090** transmits **1097** the encrypted hosted commands to a payload antenna **1080** on the vehicle **1010**. The hosted receiving antenna **1090** transmits **1097** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **1080** transmits the encrypted hosted commands to a payload **1005**. The payload **1005** on the vehicle **1010** receives the encrypted hosted commands. The payload **1005** then transmits **1055** the encrypted hosted commands to a second communication security module **1065**. The second communication security module **1065** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.

(198) It should be noted that the second communication security module **1065** may comprise one or more modules. In addition, the second communication security module **1065** may comprise one or more processors.
(199) The first communication security module **1062** then transmits **1070** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **1005**. The second communication security module **1065** transmits **1075** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **1005**. The payload **1005** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **1080** then transmits (e.g., in one or more antenna beams **1081**) payload data to the host receiving antenna **1085** and/or the hosted receiving antenna **1090** on the ground.
(200) Also, it should be noted that, although in FIG. **10**A, antenna beams **1081** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **1081** may include more or less number of beams than is shown in FIG. **10**A (e.g., antenna beams **1081** may only include a single beam), and antenna beams **1081** may include beams of different shapes than circular spot beams as is shown in FIG. **10**A (e.g., antenna beams **1081** may include elliptical beams and/or shaped beams of various different shapes).
(201) It should be noted that in one or more embodiments, the payload antenna **1080** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **1080** may comprise one or more multifeed antenna arrays.
(202) The payload **1005** transmits **1091** unencrypted host telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **1005** that is utilized by the host SOC **1050**) to the first communication security module **1062**. The first communication security module **1062** then encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry (i.e. encrypted host TLM).
(203) The payload **1005** transmits **1092** unencrypted hosted telemetry (i.e. unencrypted HoP TLM, which is telemetry data related to the portion of the payload **1005** that is leased by the HOC **1060**) to the second communication security module **1065**. The second communication security module **1065** then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry (i.e. encrypted HoP TLM).
(204) The first communication security module **1062** then transmits **1093** the encrypted host telemetry to the payload **1005**. The payload **1005** then transmits the encrypted host telemetry to the payload antenna **1080**. The payload antenna **1080** transmits **1095** the encrypted host telemetry to the host receiving antenna **1085**. The payload antenna **1080** transmits **1095** the encrypted host telemetry utilizing an in-band frequency band(s). The host receiving antenna **1085** then transmits **1094** the encrypted host telemetry to the host SOC **1050**. The host SOC **1050** then decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry.
(205) The second communication security module **1065** then transmits **1096** the encrypted hosted telemetry to the payload **1005**. The payload **1005** then transmits the encrypted hosted telemetry to the payload antenna **1080**. The payload antenna **1080** transmits **1098** the encrypted hosted telemetry to the hosted receiving antenna **1090**. The payload antenna **1080** transmits **1098** the encrypted hosted telemetry utilizing an in-band frequency band(s). The hosted receiving antenna **1090** then transmits **1099** the encrypted hosted telemetry to the HOC **1060**. The HOC **1060** then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry.
(206) FIG. **10**B is a diagram **10000** showing the disclosed system for a virtual transponder utilizing inband telemetry and commanding for the host user (i.e. host SOC) **10050** and the hosted user (i.e. HOC) **10060** using one receiving antenna and employing two COMSEC varieties for telemetry, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **10010**,

a host SOC **10050**, and a HOC **10060** are shown. The HOC **10060** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **10005** of the vehicle **10010** from the owner of a satellite (i.e. the host SOC) **10050**. It should be noted that in some embodiments, the HOC **10060** may lease all of the payload **10005** of the vehicle **10010** from the owner of a satellite (i.e. the host SOC) **10050**. Also, it should be noted that is some embodiments, the HOC **10060** may own the payload **10005** (e.g., a steerable antenna) of the vehicle **10010**, and contract the host SOC **10050** to transmit encrypted hosted commands to the vehicle **10010**.

(207) During operation, the HOC **10060** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **10005** that the HOC **10060** is leasing from the host SOC **10050**. The host SOC **10050** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **10005** that host SOC **10050** is utilizing for itself.

(208) It should be noted that, although in FIG. **10**B the host SOC **10050** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **10050** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).

(209) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).

(210) The host SOC **10050** then transmits **10020** the encrypted host commands to a host receiving antenna **10085**. After the host receiving antenna **10085** receives the encrypted host commands, the host receiving antenna **10085** transmits **10021** the encrypted host commands to a payload antenna **10080** on the vehicle **10010**. The host receiving antenna **10085** transmits **10021** the encrypted host commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **10080** transmits the encrypted host commands to a payload **10005**. The payload **10005** on the vehicle **10010** receives the encrypted host commands. The payload **10005** then transmits **10052** the encrypted host commands to a first communication security module **10062**. The first communication security module **10062** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.

(211) It should be noted that the first communication security module **10062** may comprise one or more modules. In addition, the first communication security module **10062** may comprise one or more processors.

(212) The HOC **10060** then transmits **10015** the encrypted hosted commands to the host SOC **10050**. The host SOC **10050** then transmits **10087** the encrypted hosted commands to the host receiving antenna **10085**. After the host receiving antenna **10085** receives the encrypted hosted commands, the host receiving antenna **10085** transmits **10097** the encrypted hosted commands to a payload antenna **10080** on the vehicle **10010**. The host receiving antenna **10085** transmits **10097** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **10080** transmits the encrypted hosted commands to a payload **10005**. The payload **10005** on the vehicle **10010** receives the encrypted hosted commands. The payload **10005** then transmits **10055** the encrypted hosted commands to a second communication security module **10065**. The second communication security module **10065** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.

(213) It should be noted that the second communication security module **10065** may comprise one or more modules. In addition, the second communication security module **10065** may comprise one or more processors.
(214) The first communication security module **10062** then transmits **10070** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **10005**. The second communication security module **10065** transmits **10075** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **10005**. The payload **10005** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **10080** then transmits (e.g., in one or more antenna beams **10081**) payload data to the host receiving antenna **10085** and/or the hosted receiving antenna **10090** on the ground.
(215) Also, it should be noted that, although in FIG. **10**B, antenna beams **10081** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **10081** may include more or less number of beams than is shown in FIG. **10**B (e.g., antenna beams **10081** may only include a single beam), and antenna beams **10081** may include beams of different shapes than circular spot beams as is shown in FIG. **10**B (e.g., antenna beams **10081** may include elliptical beams and/or shaped beams of various different shapes).
(216) It should be noted that in one or more embodiments, the payload antenna **10080** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **10080** may comprise one or more multifeed antenna arrays.
(217) The payload **10005** transmits **10091** unencrypted host telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **10005** that is utilized by the host SOC **10050**) to the first communication security module **10062**. The first communication security module **10062** then encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry (i.e. encrypted host TLM).
(218) The payload **10005** transmits **10092** unencrypted hosted telemetry (i.e. unencrypted HoP TLM, which is telemetry data related to the portion of the payload **10005** that is leased by the HOC **10060**) to the second communication security module **10065**. The second communication security module **10065** then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry (i.e. encrypted HoP TLM).
(219) The first communication security module **10062** then transmits **10093** the encrypted host telemetry to the payload **10005**. The payload **10005** then transmits the encrypted host telemetry to the payload antenna **10080**. The payload antenna **10080** transmits **10095** the encrypted host telemetry to the host receiving antenna **10085**. The payload antenna **10080** transmits **10095** the encrypted host telemetry utilizing an in-band frequency band(s). The host receiving antenna **10085** then transmits **10094** the encrypted host telemetry to the host SOC **10050**. The host SOC **10050** then decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry.
(220) The second communication security module **10065** then transmits **10096** the encrypted hosted telemetry to the payload **10005**. The payload **10005** then transmits the encrypted hosted telemetry to the payload antenna **10080**. The payload antenna **10080** transmits **10098** the encrypted hosted telemetry to the host receiving antenna **10085**. The payload antenna **10080** transmits **10098** the encrypted hosted telemetry utilizing an in-band frequency band(s). The host receiving antenna **10085** then transmits **10086** the encrypted hosted telemetry to the host SOC **10050**. The host SOC **10050** then transmits **10099** the encrypted hosted telemetry to the HOC **10060**. The HOC **10060** then decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry.
(221) FIGS. **11**A, **11**B, **11**C, and **11**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using two receiving antennas and employing two COMSEC varieties for telemetry, in

accordance with at least one embodiment of the present disclosure. At the start **1100** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **1105**. Then, the HOC transmits the encrypted hosted commands to a hosted receiving antenna **1110**. The hosted receiving antenna then transmits the encrypted hosted commands to a payload antenna on a vehicle **1115**. Then, the payload antenna transmits the encrypted hosted commands to a payload on the vehicle **1120**. The payload then transmits the encrypted hosted commands to a second communication security module **1125**.

(222) Then, a host spacecraft operations center (SOC) encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **1130**. The host SOC then transmits the encrypted host commands to a host receiving antenna **1135**. Then, the host receiving antenna transmits the encrypted host commands to the payload antenna **1140**. The payload antenna then transmits the encrypted host commands to the payload **1145**. Then, the payload transmits the encrypted host commands to a first communication security module **1150**. The first communication security module then decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **1155**. Then, the second communication security module decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **1160**. Then, the first communication security module transmits the unencrypted host commands to the payload **1165**. The second communication security module then transmits the unencrypted hosted commands to the payload **1170**.

(223) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **1175**. The payload antenna then transmits payload data to the host receiving antenna and/or the hosted receiving antenna **1180**. Then, the payload transmits to the first communication security module unencrypted host telemetry **1185**. The payload then transmits to the second communication security module unencrypted hosted telemetry **1190**. Then, the first communication security module encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry **1191**. The second communication security module then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry **1192**.

(224) Then, the first communication security module transmits the encrypted host telemetry to the payload **1193**. The payload then transmits the encrypted host telemetry to the payload antenna **1194**. Then, the payload antenna transmits the encrypted host telemetry to the host receiving antenna **1195**. The host receiving antenna transmits the encrypted host telemetry to the host SOC **1196**. Then, the host SOC decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **1197**.

(225) Then, the second communication security module transmits the encrypted hosted telemetry to the payload **1198**. The payload then transmits the encrypted hosted telemetry to the payload antenna **1199**. Then, the payload antenna transmits the encrypted hosted telemetry to the hosted receiving antenna **1101**. The hosted receiving antenna transmits the encrypted hosted telemetry to the HOC **1102**. Then, the HOC decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry **1103**. Then, the method ends **1104**.

(226) FIGS. **11**E, **11**F, **11**G, and **11**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using one receiving antenna and employing two COMSEC varieties for telemetry, in accordance with at least one embodiment of the present disclosure. At the start **11000** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **11005**. Then, the HOC transmits the encrypted hosted commands to a host spacecraft operations center (SOC) **11010**. The host SOC then transmits the encrypted hosted commands to a host receiving antenna **11012**. The host receiving antenna then transmits the encrypted hosted

commands to a payload antenna on a vehicle **11015**. Then, the payload antenna transmits the encrypted hosted commands to a payload on the vehicle **11020**. The payload then transmits the encrypted hosted commands to a second communication security module **11025**.
(227) Then, a host spacecraft operations center (SOC) encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **11030**. The host SOC then transmits the encrypted host commands to a host receiving antenna **11035**. Then, the host receiving antenna transmits the encrypted host commands to the payload antenna **11040**. The payload antenna then transmits the encrypted host commands to the payload **11045**. Then, the payload transmits the encrypted host commands to a first communication security module **11050**. The first communication security module then decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **11055**. Then, the second communication security module decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **11060**. Then, the first communication security module transmits the unencrypted host commands to the payload **11065**. The second communication security module then transmits the unencrypted hosted commands to the payload **11070**.
(228) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **11075**. The payload antenna then transmits payload data to the host receiving antenna and/or the hosted receiving antenna **11080**. Then, the payload transmits to the first communication security module unencrypted host telemetry **11085**. The payload then transmits to the second communication security module unencrypted hosted telemetry **11090**. Then, the first communication security module encrypts the unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry **11091**. The second communication security module then encrypts the unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry **11092**.
(229) Then, the first communication security module transmits the encrypted host telemetry to the payload **11093**. The payload then transmits the encrypted host telemetry to the payload antenna **11094**. Then, the payload antenna transmits the encrypted host telemetry to the host receiving antenna **11095**. The host receiving antenna transmits the encrypted host telemetry to the host SOC **11096**. Then, the host SOC decrypts the encrypted host telemetry utilizing the first COMSEC variety to generate the unencrypted host telemetry **11097**.
(230) Then, the second communication security module transmits the encrypted hosted telemetry to the payload **11098**. The payload then transmits the encrypted hosted telemetry to the payload antenna **11099**. Then, the payload antenna transmits the encrypted hosted telemetry to the host receiving antenna **11001**. The host receiving antenna transmits the encrypted hosted telemetry to the host SOC **11002**. The host SOC then transmits the encrypted hosted telemetry to the HOC **11003**. Then, the HOC decrypts the encrypted hosted telemetry utilizing the second COMSEC variety to generate the unencrypted hosted telemetry **11004**. Then, the method ends **11005**.
(231) FIG. **12**A is a diagram **1200** showing the disclosed system for a virtual transponder utilizing inband telemetry and commanding for the host user (i.e. host SOC) **1250** and the hosted user (i.e. HOC) **1260** using two receiving antennas and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **1210**, a host SOC **1250**, and a HOC **1260** are shown. The HOC **1260** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **1205** of the vehicle **1210** from the owner of a satellite (i.e. the host SOC) **1250**. It should be noted that in some embodiments, the HOC **1260** may lease all of the payload **1205** of the vehicle **1210** from the owner of a satellite (i.e. the host SOC) **1250**. Also, it should be noted that is some embodiments, the HOC **1260** may own the payload **1205** (e.g., a steerable antenna) of the vehicle **1210**, and contract the host SOC **1250** to transmit encrypted hosted commands to the vehicle **1210**.
(232) During operation, the HOC **1260** encrypts unencrypted hosted commands (i.e. unencrypted

HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e. encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **1205** that the HOC **1260** is leasing from the host SOC **1250**. The host SOC **1250** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **1205** that host SOC **1250** is utilizing for itself.
(233) It should be noted that, although in FIG. **12**A the host SOC **1250** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **1250** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).
(234) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).
(235) The host SOC **1250** then transmits **1220** the encrypted host commands to a host receiving antenna **1285**. After the host receiving antenna **1285** receives the encrypted host commands, the host receiving antenna **1285** transmits **1221** the encrypted host commands to a payload antenna **1280** on the vehicle **1210**. The host receiving antenna **1285** transmits **1221** the encrypted host commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **1280** transmits the encrypted host commands to a payload **1205**. The payload **1205** on the vehicle **1210** receives the encrypted host commands. The payload **1205** then transmits **1252** the encrypted host commands to a first communication security module **1262**. The first communication security module **1262** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.
(236) It should be noted that the first communication security module **1262** may comprise one or more modules. In addition, the first communication security module **1262** may comprise one or more processors.
(237) The HOC **1260** then transmits **1215** the encrypted hosted commands to a hosted receiving antenna **1290**. After the hosted receiving antenna **1290** receives the encrypted hosted commands, the hosted receiving antenna **1290** transmits **1297** the encrypted hosted commands to a payload antenna **1280** on the vehicle **1210**. The hosted receiving antenna **1290** transmits **1297** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **1280** transmits the encrypted hosted commands to a payload **1205**. The payload **1205** on the vehicle **1210** receives the encrypted hosted commands. The payload **1205** then transmits **1255** the encrypted hosted commands to a second communication security module **1265**. The second communication security module **1265** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.
(238) It should be noted that the second communication security module **1265** may comprise one or more modules. In addition, the second communication security module **1265** may comprise one or more processors.
(239) The first communication security module **1262** then transmits **1270** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **1205**. The second communication security module **1265** transmits **1275** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **1205**. The payload **1205** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **1280** then transmits (e.g., in one or more antenna beams **1281**) payload data to the host receiving antenna

1285 and/or the hosted receiving antenna **1290** on the ground.

(240) Also, it should be noted that, although in FIG. **12**A, antenna beams **1281** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **1281** may include more or less number of beams than is shown in FIG. **12**A (e.g., antenna beams **1281** may only include a single beam), and antenna beams **1281** may include beams of different shapes than circular spot beams as is shown in FIG. **12**A (e.g., antenna beams **1281** may include elliptical beams and/or shaped beams of various different shapes).

(241) It should be noted that in one or more embodiments, the payload antenna **1280** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **1280** may comprise one or more multifeed antenna arrays.

(242) The payload **1205** transmits **1291** unencrypted telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **1205** that is utilized by the host SOC **1250**; and unencrypted HoP TLM, which is telemetry data related to the portion of the payload **1205** that is leased by the HOC **1260**) to the first communication security module **1262**. The first communication security module **1262** then encrypts the unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry (i.e. encrypted TLM). The first communication security module **1262** then transmits **1293** the encrypted telemetry to the payload **1205**. The payload **1205** then transmits the encrypted telemetry to the payload antenna **1280**.

(243) The payload antenna **1280** transmits **1295** the encrypted telemetry to the host receiving antenna **1285**. The payload antenna **1280** transmits **1295** the encrypted telemetry utilizing an in-band frequency band(s). The host receiving antenna **1285** then transmits **1294** the encrypted telemetry to the host SOC **1250**. The host SOC **1250** then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry. The host SOC **1250** then utilizes a database that comprises host payload decommutated information and does not comprise hosted payload decommutated information (i.e. a database without hosted payload decommutated information) to read to unencrypted telemetry to determine the telemetry data related to the portion of the payload **1205** that is utilized by the host SOC **1250**.

(244) The payload antenna **1280** transmits **1298** the encrypted telemetry to the hosted receiving antenna **1290**. The payload antenna **1280** transmits **1298** the encrypted telemetry utilizing an in-band frequency band(s). The hosted receiving antenna **1290** then transmits **1299** the encrypted telemetry to the HOC **1260**. The HOC **1260** then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry. The HOC **1260** then utilizes a database that comprises hosted payload decommutated information and does not comprise host payload decommutated information (i.e. a database without host payload decommutated information) to read to unencrypted telemetry to determine the telemetry data related to the portion of the payload **1205** that is utilized by the HOC **1260**.

(245) FIG. **12**B is a diagram **12000** showing the disclosed system for a virtual transponder utilizing inband telemetry and commanding for the host user (i.e. host SOC) **12050** and the hosted user (i.e. HOC) **12060** using one receiving antenna and employing one COMSEC variety for telemetry, in accordance with at least one embodiment of the present disclosure. In this figure, a vehicle **12010**, a host SOC **12050**, and a HOC **12060** are shown. The HOC **12060** has leased at least a portion (e.g., a virtual transponder(s)) of the payload **12005** of the vehicle **12010** from the owner of a satellite (i.e. the host SOC) **12050**. It should be noted that in some embodiments, the HOC **12060** may lease all of the payload **12005** of the vehicle **12010** from the owner of a satellite (i.e. the host SOC) **12050**. Also, it should be noted that is some embodiments, the HOC **12060** may own the payload **12005** (e.g., a steerable antenna) of the vehicle **12010**, and contract the host SOC **12050** to transmit encrypted hosted commands to the vehicle **12010**.

(246) During operation, the HOC **12060** encrypts unencrypted hosted commands (i.e. unencrypted HoP CMD), by utilizing a second COMSEC variety, to produce encrypted hosted commands (i.e.

encrypted HoP CMD). The hosted commands are commands that are used to configure the portion (e.g., a virtual transponder(s)) of the payload **12005** that the HOC **12060** is leasing from the host SOC **12050**. The host SOC **12050** encrypts unencrypted host commands (i.e. unencrypted host CMD), by utilizing a first COMSEC variety, to produce encrypted host commands (i.e. encrypted host CMD). The host commands are commands that are used to configure the portion (e.g., a transponder(s)) of the payload **12005** that host SOC **12050** is utilizing for itself.
(247) It should be noted that, although in FIG. **12**B the host SOC **12050** is depicted to have its ground antenna located right next to its operations building; in other embodiments, the host SOC **12050** may have its ground antenna located very far away from the its operations building (e.g., the ground antenna may be located in another country than the operations building).
(248) Also, it should be noted that the first COMSEC variety may include at least one encryption key and/or at least one algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm). Additionally, it should be noted that the second COMSEC variety may include at least one encryption key and/or at least one encryption algorithm (e.g., a Type 1 encryption algorithm or a Type 2 encryption algorithm).
(249) The host SOC **12050** then transmits **12020** the encrypted host commands to a host receiving antenna **12085**. After the host receiving antenna **12085** receives the encrypted host commands, the host receiving antenna **12085** transmits **12021** the encrypted host commands to a payload antenna **12080** on the vehicle **12010**. The host receiving antenna **12085** transmits **12021** the encrypted host commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **12080** transmits the encrypted host commands to a payload **12005**. The payload **12005** on the vehicle **12010** receives the encrypted host commands. The payload **12005** then transmits **12052** the encrypted host commands to a first communication security module **12062**. The first communication security module **12062** decrypts the encrypted host commands utilizing the first COMSEC variety (i.e. COMSEC Variety 1) to generate unencrypted host commands.
(250) It should be noted that the first communication security module **12062** may comprise one or more modules. In addition, the first communication security module **12062** may comprise one or more processors.
(251) The HOC **12060** then transmits **12015** the encrypted hosted commands to the host SOC **12050**. The host SOC **12050** then transmits **12098** the encrypted hosted commands to the host receiving antenna **12085**. After the host receiving antenna **12085** receives the encrypted hosted commands, the host receiving antenna **12085** transmits **12097** the encrypted hosted commands to a payload antenna **12080** on the vehicle **12010**. The host receiving antenna **12085** transmits **12097** the encrypted hosted commands utilizing an in-band frequency band(s) (i.e. a frequency band(s) that is the same frequency band(s) utilized to transmit payload data). The payload antenna **12080** transmits the encrypted hosted commands to a payload **12005**. The payload **12005** on the vehicle **12010** receives the encrypted hosted commands. The payload **12005** then transmits **12055** the encrypted hosted commands to a second communication security module **12065**. The second communication security module **12065** decrypts the encrypted hosted commands utilizing the second COMSEC variety (i.e. COMSEC Variety 2) to generate unencrypted hosted commands.
(252) It should be noted that the second communication security module **12065** may comprise one or more modules. In addition, the second communication security module **12065** may comprise one or more processors.
(253) The first communication security module **12062** then transmits **12070** the unencrypted host commands to the payload (i.e. the shared host/hosted payload) **12005**. The second communication security module **12065** transmits **12075** the unencrypted hosted commands to the payload (i.e. the shared host/hosted payload) **12005**. The payload **12005** is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands. The payload antenna **12080** then transmits (e.g., in one or more antenna beams **12081**) payload data to the host receiving

antenna **12085** and/or the hosted receiving antenna **12090** on the ground.

(254) Also, it should be noted that, although in FIG. **12**B, antenna beams **12081** is shown to include a plurality of circular spot beams; in other embodiments, antenna beams **12081** may include more or less number of beams than is shown in FIG. **12**B (e.g., antenna beams **12081** may only include a single beam), and antenna beams **12081** may include beams of different shapes than circular spot beams as is shown in FIG. **12**B (e.g., antenna beams **12081** may include elliptical beams and/or shaped beams of various different shapes).

(255) It should be noted that in one or more embodiments, the payload antenna **12080** may comprise one or more reflector dishes including, but not limited to, parabolic reflectors and/or shaped reflectors. In some embodiments, the payload antenna **12080** may comprise one or more multifeed antenna arrays.

(256) The payload **12005** transmits **12091** unencrypted telemetry (i.e. unencrypted host TLM, which is telemetry data related to the portion of the payload **12005** that is utilized by the host SOC **12050**; and unencrypted HoP TLM, which is telemetry data related to the portion of the payload **12005** that is leased by the HOC **12060**) to the first communication security module **12062**. The first communication security module **12062** then encrypts the unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry (i.e. encrypted TLM). The first communication security module **12062** then transmits **12093** the encrypted telemetry to the payload **12005**. The payload **12005** then transmits the encrypted telemetry to the payload antenna **12080**.

(257) The payload antenna **12080** transmits **12095** the encrypted telemetry to the host receiving antenna **12085**. The payload antenna **12080** transmits **12095** the encrypted telemetry utilizing an in-band frequency band(s). The host receiving antenna **12085** then transmits **12094** the encrypted telemetry to the host SOC **12050**. The host SOC **12050** then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry. The host SOC **12050** then utilizes a database that comprises host payload decommutated information and does not comprise hosted payload decommutated information (i.e. a database without hosted payload decommutated information) to read to unencrypted telemetry to determine the telemetry data related to the portion of the payload **12005** that is utilized by the host SOC **12050**.

(258) The host SOC **12050** then transmits **12099** the encrypted telemetry to the HOC **12060**. The HOC **12060** then decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry. The HOC **12060** then utilizes a database that comprises hosted payload decommutated information and does not comprise host payload decommutated information (i.e. a database without host payload decommutated information) to read to unencrypted telemetry to determine the telemetry data related to the portion of the payload **12005** that is utilized by the HOC **12060**.

(259) FIGS. **13**A, **13**B, **13**C, and **13**D together show a flow chart for the disclosed method for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using two receiving antennas and employing one COMSEC variety, in accordance with at least one embodiment of the present disclosure. At the start **1300** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **1305**. Then, the HOC transmits the encrypted hosted commands to a hosted receiving antenna **1310**. The hosted receiving antenna then transmits the encrypted hosted commands to a payload antenna on a vehicle **1315**. Then, the payload antenna transmits the encrypted hosted commands to a payload on the vehicle **1320**. The payload then transmits the encrypted hosted commands to a second communication security module **1325**.

(260) Then, a host spacecraft operations center (SOC) encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **1330**. The host SOC then transmits the encrypted host commands to a host receiving antenna **1335**. Then, the host receiving antenna transmits the encrypted host commands to the payload antenna **1340**. Then, the payload

antenna transmits the encrypted host commands to the payload **1345**. The payload then transmits the encrypted host commands to a first communication security module **1350**. Then, the first communication security module decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **1355**. The second communication security module then decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **1360**. Then, the first communication security module transmits the unencrypted host commands to the payload **1365**. The second communication security module then transmits the unencrypted hosted commands to the payload **1370**.

(261) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **1375**. The payload antenna then transmits payload data to the host receiving antenna and/or the hosted receiving antenna **1380**. Then, the payload transmits to the first communication security module unencrypted telemetry **1385**. The first communication security module then encrypts the unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry **1390**.

(262) Then, the first communication security module transmits the encrypted telemetry to the payload **1391**. The payload then transmits the encrypted telemetry to the payload antenna **1392**. Then, the payload antenna transmits the encrypted telemetry to the host receiving antenna **1393**. The host receiving antenna then transmits the encrypted telemetry to the host SOC **1394**. Then, the host SOC decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry **1395**. The host SOC then determines telemetry data related to a portion of the payload utilized by the host SOC by using a database without hosted decommutated information to read the unencrypted telemetry **1396**.

(263) Then, the payload antenna transmits the encrypted telemetry to the hosted receiving antenna **1397**. The hosted receiving antenna then transmits the encrypted telemetry to the HOC **1398**. Then, the HOC decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry **1399**. The HOC then determines telemetry data related to a portion of the payload utilized by the HOC by using a database without host decommutated information to read the unencrypted telemetry **1301**. Then, the method ends **1302**.

(264) FIGS. **13**E, **13**F, **13**G, and **13**H together show a flow chart for the disclosed method for a virtual transponder utilizing inband telemetry and commanding for the host user and the hosted user using one receiving antenna and employing one COMSEC variety, in accordance with at least one embodiment of the present disclosure. At the start **13000** of the method, a hosted payload (HoP) operation center (HOC) encrypts unencrypted hosted commands by utilizing a second communication security (COMSEC) variety to produce encrypted hosted commands **13005**. Then, the HOC transmits the encrypted hosted commands to a host spacecraft operations center (SOC) **13010**. The host SOC then transmits the encrypted hosted commands to a host receiving antenna **13012**. The host receiving antenna then transmits the encrypted hosted commands to a payload antenna on a vehicle **13015**. Then, the payload antenna transmits the encrypted hosted commands to a payload on the vehicle **13020**. The payload then transmits the encrypted hosted commands to a second communication security module **13025**.

(265) Then, a host spacecraft operations center (SOC) encrypts unencrypted host commands by utilizing a first COMSEC variety to produce encrypted host commands **13030**. The host SOC then transmits the encrypted host commands to a host receiving antenna **13035**. Then, the host receiving antenna transmits the encrypted host commands to the payload antenna **13040**. Then, the payload antenna transmits the encrypted host commands to the payload **13045**. The payload then transmits the encrypted host commands to a first communication security module **13050**. Then, the first communication security module decrypts the encrypted host commands utilizing the first COMSEC variety to generate the unencrypted host commands **13055**. The second communication security module then decrypts the encrypted hosted commands utilizing the second COMSEC variety to generate the unencrypted hosted commands **13060**. Then, the first communication

security module transmits the unencrypted host commands to the payload **13065**. The second communication security module then transmits the unencrypted hosted commands to the payload **13070**.

(266) Then, the payload is reconfigured according to the unencrypted host commands and/or the unencrypted hosted commands **13075**. The payload antenna then transmits payload data to the host receiving antenna and/or the hosted receiving antenna **13080**. Then, the payload transmits to the first communication security module unencrypted telemetry **13085**. The first communication security module then encrypts the unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry **13090**.

(267) Then, the first communication security module transmits the encrypted telemetry to the payload **13091**. The payload then transmits the encrypted telemetry to the payload antenna **13092**. Then, the payload antenna transmits the encrypted telemetry to the host receiving antenna **13093**. The host receiving antenna then transmits the encrypted telemetry to the host SOC **13094**. Then, the host SOC decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry **13095**. The host SOC then determines telemetry data related to a portion of the payload utilized by the host SOC by using a database without hosted decommutated information to read the unencrypted telemetry **13096**.

(268) The host SOC then transmits the encrypted telemetry to the HOC **13098**. Then, the HOC decrypts the encrypted telemetry utilizing the first COMSEC variety to generate the unencrypted telemetry **13099**. The HOC then determines telemetry data related to a portion of the payload utilized by the HOC by using a database without host decommutated information to read the unencrypted telemetry **13001**. Then, the method ends **13002**.

(269) FIGS. **14-19**B show exemplary systems and methods for a virtual transponder, in accordance with at least one embodiment of the present disclosure.

(270) FIG. **14** is a diagram **1400** showing the disclosed system for a virtual transponder on a vehicle **1610**, in accordance with at least one embodiment of the present disclosure. In this figure, a computing device **1410** is shown. The computing device **1410** may be located at a station (e.g., a host station or a hosted station). When the computing device **1410** is located at a host station (i.e. a station operated by a host user (Host SOC)), the computing device **1410** is referred to as a host computing device. And, when the computing device **1410** is located at a hosted station (i.e. a station operated by a hosted user (HOC)), the computing device **1410** is referred to as a hosted computing device. In one or more embodiments, the station is a ground station **1415**, a terrestrial vehicle (e.g., a military jeep) **1420**, an airborne vehicle (e.g., an aircraft) **1425**, or a marine vehicle (e.g., a ship) **1430**.

(271) During operation, a user (e.g., a host user or a hosted user) **1405** selects, via a graphical user interface (GUI) (e.g., a host GUI or a hosted GUI) **1435** displayed on a screen of the computing device **1410** (e.g., a host computing device or a hosted computing device), an option (e.g., a value) for each of at least one different variable for a portion of the payload **1680** on the vehicle **1610** utilized by the user **1405**. It should be noted that the details of payload **1680** as is illustrated in FIG. **16** is depicted on the GUI **1435**, which is displayed on the screen of the computing device **1410**.

(272) Refer FIG. **16** to view the different variables of the payload **1680** on the vehicle **1610** that may be selected by the user **1405** by using the GUI **1435** that is displayed to the user **1405**. Also, refer to FIG. **17** to view the different variables of the digital channelizer **1670** of the payload **1680** that may be selected by the user **1405** by using the GUI **1435** that is displayed to the user **1405**. In one or more embodiments, various different variables may be presented by the GUI **1435** to be selected including, but not limited to, at least one transponder power, at least one transponder spectrum, at least one transponder gain setting, at least one transponder limiter setting, at least one transponder automatic level control setting, at least one transponder phase setting, at least one internal gain generation, bandwidth for at least one beam, at least one frequency band for at least one beam, at least one transponder beamforming setting, effective isotropic radiation power (EIRP)

for at least one beam, at least one transponder channel, and/or beam steering for at least one beam. It should be noted that the user **1405** may select an option by clicking on the associated variable (e.g., clicking on one of the mixers **1665** to change the frequency band of the mixer's associated transmit antenna **1655**) in the payload **1680** by using the GUI **1435**, and by either typing in a value or selecting a value from a drop down menu (e.g., by typing in a desired transmission frequency band for the associated transmit antenna **1655**). It should be noted that the payload **1680** depicted in FIG. **16** is an exemplary payload, and the depiction does not show all possible different variables that may be selected by user **1405** by using the GUI **1435**.

(273) After the user **1405** has selected, via the GUI **1435** displayed on the computing device **1410**, an option for each of at least one variable for the portion of the payload **1680** on the vehicle **1610** utilized by the user **1405**, the option(s) is transmitted **1440** to a configuration algorithm (CA) **1445** (e.g., an algorithm contained in an XML file, such as CAConfig.xml **1450**). The CA **1445** then generates a configuration for the portion of the payload **1680** on the vehicle **1610** utilized by the user **1405** by using the option(s). Then, the CA **1445** transmits **1455** the configuration to a command generator (e.g., a host command generator or a hosted command generator) **1460**. Optionally, the CA **1445** also stores the configuration in a report file **1465**.

(274) After the command generator **1460** has received the configuration, the command generator **1460** generates commands (e.g., host commands or hosted commands) for reconfiguring the portion of the payload **1680** on the vehicle **1610** utilized by the user **1405** by using the configuration. Then, the commands are transmitted **1470** to an encryption module **1475**. After receiving the commands, the encryption module **1475** then encrypts the commands (e.g., by utilizing a first COMSEC variety or a second COMSEC variety) to generate encrypted commands (e.g., host encrypted commands or hosted encrypted commands).

(275) Then, the encrypted commands are transmitted **1480** from the station (e.g., a ground station **1415**, a terrestrial vehicle (e.g., a military jeep) **1420**, an airborne vehicle (e.g., an aircraft) **1425**, or a marine vehicle (e.g., a ship) **1430**) to the vehicle **1410**. It should be noted that, in one or more embodiments, the computing device **1410**, the CA **1445**, the command generator **1460**, and the encryption module **1475** are all located at the station (e.g., the host station or the hosted station). In other embodiments, some or more of these items may be located in different locations. In addition, in one or more embodiments, the vehicle **1610** is an airborne vehicle (e.g., a satellite, an aircraft, an unmanned vehicle (UAV), or a space plane).

(276) After the vehicle **1610** has received the encrypted commands, the vehicle decrypts the commands to generated unencrypted commands (e.g., host unencrypted commands or hosted unencrypted commands). Then, the portion of the payload **1680** on the vehicle **1610** utilized by the user **1405** is reconfigured by using the unencrypted commands. In one or more embodiments, the reconfiguring of the payload **1680** may comprise reconfiguring at least one antenna **1615**, **1655** (refer to FIG. **16**), at least one analog-to-digital converter, at least one digital-to-analog converter, at least one beamformer, at least one digital channelizer **1710** (refer to FIG. **17**), at least one demodulator, at least one modulator, at least one digital switch matrix **1720** (refer to FIG. **17**), and/or at least one digital combiner **1730** (refer to FIG. **17**). It should be noted that in other embodiments, the reconfiguring of the payload **1680** may comprise reconfiguring at least one analog switch matrix.

(277) FIG. **15** is a diagram **1500** showing an exemplary allocation of bandwidth amongst a plurality of beams (U**1**-U**45**) when utilizing the disclosed virtual transponder, in accordance with at least one embodiment of the present disclosure. In this figure, the bandwidth of each of the beams (U**1**-U**45**) is illustrated as a bar.

(278) On the left side **1510** of the diagram **1500**, a portion of the bandwidth of each of the beams (U**1**-U**45**) is shown to be utilized by only the host user (i.e. the owner of the vehicle). For this example, the host user is not leasing out any portion of the payload to a hosted user (i.e. a customer).

(279) On the right side **1520** of the diagram **1500**, a portion of the bandwidth of each of the beams is shown to be utilized by the host user (i.e. the owner of the vehicle). Also, at least some (if not all) of the portion of the bandwidth of each of the beams (U**1**-U**45**) not utilized by the host user, is shown to be utilized by the hosted user (i.e. a customer). For this example, the host user is leasing out a portion of the payload to a hosted user (i.e. a customer). Specifically, the host user is leasing out a portion the bandwidth of each of the beams (U**1**-U**45**) to the hosted user.
(280) It should be noted that in other embodiments, the host user may lease out the entire bandwidth of some (if not all) of beam(s) to the hosted user. For these, embodiments, the hosted user alone will utilize the bandwidth of these leased beam(s).
(281) FIG. **16** is a diagram **1600** showing the switch architecture for a flexible allocation of bandwidth amongst a plurality of beams (U**1**-UN) (i.e. including uplink and downlink beams) when utilizing the disclosed virtual transponder, in accordance with at least one embodiment of the present disclosure. In this figure, details of a payload **1680** on a vehicle **1610** are shown. In particular, each of a plurality (i.e. N number) of receive antennas **1615**, on the vehicle **1610**, is shown to be receiving one of the uplink beams (U**1**-UN). As such, for example, receive antenna **1615** connected to input port **1** receives uplink beam U**6**, receive antenna **1615** connected to input port **2** receives uplink beam U**14**, and receive antenna **1615** connected to input port N receives uplink beam U**34**. Each receive antenna **1615** is shown to be followed by a polarizer (i.e. pol) **1620** and a waveguide filter (i.e. WG Filter) **1625**.
(282) Also, in this figure, each of a plurality (i.e. N number) of transmit antennas **1655**, on the vehicle **1610**, is shown to be receiving one of the downlink beams (U**1**-UN). As such, for example, transmit antenna **1655** connected to output port **1** receives downlink beam U**19**, transmit antenna **1655** connected to output port **2** receives downlink beam U**6**, and transmit antenna **1655** connected to output port N receives downlink beam U**1**. Each transmit antenna **1655** is shown to be preceded by a polarizer (i.e. pol) **1645** and a waveguide filter (i.e. WG Filter) **1650**.
(283) It should be noted that, in one or more embodiments, various different types of antennas may be employed for the receive antennas **1615** and the transmit antennas **1655** including, but not limited to, parabolic reflector antennas, shaped reflector antennas, multifeed array antennas, phase array antennas, and/or any combination thereof.
(284) During operation, a host user **1605** encrypts unencrypted host commands to produce encrypted host commands. Also, a hosted user **1630** encrypts unencrypted hosted commands to produce encrypted hosted commands. The hosted user **1630** transmits **1635** the encrypted hosted commands to the host user **1605**. The host user **1605** transmits **1640** the encrypted host commands and the encrypted hosted commands to the vehicle **1610**. The encrypted host commands and encrypted hosted commands are decrypted on the vehicle **1610** to produce the unencrypted host commands and unencrypted hosted commands.
(285) Then, the payload on the vehicle **1610** receives the unencrypted host commands and unencrypted hosted commands. The digital channelizer **1670** then reconfigures the channels of the uplink beams (U**1**-UN) and downlink beams (U**1**-UN) according to the unencrypted host commands and unencrypted hosted commands. The configuring of the channels allocates the bandwidth of the uplink beams (U**1**-UN) and downlink beams (U**1**-UN) amongst the host user **1605** and the hosted user **1630**.
(286) Also, the transmit antennas **1655** and the receive antennas **1615** are configured according to the unencrypted host commands and/or unencrypted hosted commands. For example, some, if not all, of the transmit antennas **1655** and/or the receive antennas **1615** may be gimbaled to project their beams on different locations on the ground. Also, for example, some, if not all, of the transmit antennas **1655** and/or the receive antennas **1615** may have their phase changed such that (1) the shape of the beam is changed (e.g., has the effect of changing the coverage area of the beam, changing the peak(s) amplitude of the beam, and/or the changing the peak(s) amplitude location on the ground), and/or (2) the beam is projected on a different location on the ground (i.e. has the

same effect as gimbaling the antenna **1615**, **1655**).
(287) Additionally, the mixers **1660** on the input ports and/or the mixers **1665** on the output ports are configured according to the unencrypted host commands and/or unencrypted hosted commands. For example, some, if not all, of the mixers **1660** on the input ports and/or the mixers **1665** on the output ports may mix in different frequency bands to change the frequency band(s) of the beams (U**1**-UN).
(288) FIG. **17** is a diagram **1700** showing details of the digital channelizer **1670** of FIG. **16**, in accordance with at least one embodiment of the present disclosure. In this figure, the digital channelizer **1670** is shown to include three main parts, which are the channelizer **1710**, the switch matrix **1720**, and the combiner **1730**. The digital channelizer **1710** divides the input beam spectrum (i.e. frequency band) from each input port into input subchannels (i.e. frequency slices). In this figure, each beam spectrum (i.e. frequency band) is shown to be divided into twelve (12) input subchannels (i.e. frequency slices). It should be noted that in other embodiments, each input beam spectrum may be divided into more or less than twelve (12) input subchannels, as is shown in FIG. **17**.
(289) The switch matrix **1720** routes the input subchannels from the input ports to their assigned respective output ports, where they are referred to as output subchannels. In this figure, five (5) exemplary types of routing that may be utilized by the switch matrix **1720** are shown, which include direct mapping **1740**, in-beam multicast **1750**, cross-beam multicast **1760**, cross-beam mapping **1770**, and cross-beam point-to-point routing **1780**. The combiner **1730** combines the output subchannels to create an output beam spectrum for each output port. As previously mentioned above, during the reconfiguring of the payload **1680**, the channelizer **1710**, the switch matrix **1720**, and/or the combiner **1730** of the digital channelizer **1670** may be reconfigured a various different number of ways (e.g., changing the dividing of the input beam spectrums into input subchannels, changing the routing of the input subchannels, and/or changing the combining of the output subchannels to create the output beam spectrums).
(290) FIG. **18** is a diagram **1800** showing exemplary components on the vehicle (e.g., satellite) **1810** that may be utilized by the disclosed virtual transponder, in accordance with at least one embodiment of the present disclosure. In this figure, various components, on the vehicle **1810**, are shown that may be configured according to the unencrypted host commands (e.g., the host channel **1830**) and/or unencrypted hosted commands (e.g., the hosted channel **1820**).
(291) In this figure, the uplink antenna **1840**, the downlink antenna **1850**, and various components of the all-digital payload **1860** (including the analog-to-digital (A/D) converter **1865**, the digital channelizer **1875**, the digital switch matrix **1895**, the digital combiner **1815**, and the digital-to-analog (D/A) converter **1835**) are shown that may be configured according to the unencrypted host commands (e.g., the host channel **1830**) and/or unencrypted hosted commands (e.g., the hosted channel **1820**). In addition, some other components of the all-digital payload **1860** (including the uplink beamforming **1870**, the demodulator **1880**, the modulator **1890**, and the downlink beamforming **1825**) may optionally be configured according to the unencrypted host commands (e.g., the host channel **1830**) and/or unencrypted hosted commands (e.g., the hosted channel **1820**).
(292) FIGS. **19**A and **19**B together show a flow chart for the disclosed method for a virtual transponder on a vehicle, in accordance with at least one embodiment of the present disclosure. At the start **1900** of the method, a host user, with a host graphical user interface (GUI) on a host computing device, selects an option for each of at least one variable for a portion of a payload on the vehicle utilized by the host user **1905**. Also, a hosted user, with a hosted GUI on a hosted computing device, selects an option for each of at least one variable for a portion of the payload on the vehicle utilized by the hosted user **1910**. Then, a configuration algorithm (CA) generates a configuration for the portion of the payload on the vehicle utilized by the host user by using the option for each of at least one variable for the portion of the payload on the vehicle utilized by the host user **1915**. Also, the CA generates a configuration for the portion of the payload on the vehicle

utilized by the hosted user by using an option for each of at least one variable for the portion of the payload on the vehicle utilized by the hosted user **1920**.

(293) A host command generator then generates host commands for reconfiguring the portion of the payload on the vehicle utilized by the host user by using the configuration for the portion of the payload on the vehicle utilized by the host user **1925**. And, a hosted command generator generates hosted commands for reconfiguring the portion of the payload on the vehicle utilized by the hosted user by using the configuration for the portion of the payload on the vehicle utilized by the hosted user **1930**. Then, the host commands and the hosted commands are transmitted to the vehicle **1935**. The portion of the payload on the vehicle utilized by the host user is then reconfigured by using the host commands **1940**. Also, the portion of the payload on the vehicle utilized by the hosted user is reconfigured by using the hosted commands **1945**. Then, the method ends **1950**.

(294) Although particular embodiments have been shown and described, it should be understood that the above discussion is not intended to limit the scope of these embodiments. While embodiments and variations of the many aspects of the invention have been disclosed and described herein, such disclosure is provided for purposes of explanation and illustration only. Thus, various changes and modifications may be made without departing from the scope of the claims.

(295) Where methods described above indicate certain events occurring in certain order, those of ordinary skill in the art having the benefit of this disclosure would recognize that the ordering may be modified and that such modifications are in accordance with the variations of the present disclosure. Additionally, parts of methods may be performed concurrently in a parallel process when possible, as well as performed sequentially. In addition, more parts or less part of the methods may be performed.

(296) Accordingly, embodiments are intended to exemplify alternatives, modifications, and equivalents that may fall within the scope of the claims.

(297) Although certain illustrative embodiments and methods have been disclosed herein, it can be apparent from the foregoing disclosure to those skilled in the art that variations and modifications of such embodiments and methods can be made without departing from the true spirit and scope of the art disclosed. Many other examples of the art disclosed exist, each differing from others in matters of detail only. Accordingly, it is intended that the art disclosed shall be limited only to the extent required by the appended claims and the rules and principles of applicable law.

Claims

1. A method for a virtual transponder utilizing inband commanding, the method comprising: receiving, by a payload antenna on a vehicle via a hosted receiving antenna, encrypted hosted commands transmitted from a hosted payload (HoP) operation center (HOC), wherein the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety; receiving, by the payload antenna via a host receiving antenna, encrypted host commands transmitted from a host spacecraft operations center (SOC), wherein the encrypted host commands are encrypted utilizing a first COMSEC variety; decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands; decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands; reconfiguring a payload on the vehicle according to at least one of the unencrypted host commands or the unencrypted hosted commands; transmitting, by the payload antenna, payload data to at least one of the host receiving antenna or the hosted receiving antenna; encrypting, by the first communication security module, unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry; transmitting, by the payload antenna via the host receiving antenna, the encrypted telemetry to the host SOC; and transmitting, by the payload antenna via the hosted

receiving antenna, the encrypted telemetry to the HOC.

2. The method of claim 1, wherein the reconfiguring of the payload according to at least one of the unencrypted host commands or the unencrypted hosted commands comprises adjusting at least one of: transponder power, transponder spectrum monitoring, transponder connectivity, transponder gain settings, transponder limiter settings, transponder automatic level control settings, transponder phase settings, internal gain generation, bandwidth for at least one beam, at least one frequency band for at least one of the at least one beam, transponder beamforming settings, effective isotropic radiation power (EIRP) for at least one of the at least one beam, transponder channels, or beam steering.

3. The method of claim 1, wherein the reconfiguring of the payload according to at least one of the unencrypted host commands or the unencrypted hosted commands comprises reconfiguring at least one of: at least one antenna, at least one analog-to-digital converter, at least one digital-to-analog converter, at least one beamformer, at least one digital channelizer, at least one demodulator, at least one modulator, at least one digital switch matrix, at least one digital combiner, or at least one analog switch matrix.

4. The method of claim 1, wherein the vehicle is an airborne vehicle.

5. The method of claim 4, wherein the airborne vehicle is one of a satellite, aircraft, unmanned aerial vehicle (UAV), or space plane.

6. A method for a virtual transponder utilizing inband commanding, the method comprising: receiving, by a payload antenna on a vehicle via a host receiving antenna, encrypted host commands and encrypted hosted commands transmitted from a host spacecraft operations center (SOC), the encrypted hosted commands transmitted to the host SOC from a hosted payload (HoP) operation center (HOC), wherein the encrypted host commands are encrypted utilizing a first communication security (COMSEC) variety, wherein the encrypted hosted commands are encrypted utilizing a second COMSEC variety; decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands; decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands; reconfiguring a payload on the vehicle according to at least one of the unencrypted host commands or the unencrypted hosted commands; transmitting, by the payload antenna, payload data to at least one of the host receiving antenna or a hosted receiving antenna; encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry; encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry; transmitting, by a host telemetry transmitter, the encrypted host telemetry to the host SOC; and transmitting, by a hosted telemetry transmitter via the host SOC, the encrypted hosted telemetry to the HOC.

7. The method of claim 6, wherein the reconfiguring of the payload according to at least one of the unencrypted host commands or the unencrypted hosted commands comprises adjusting at least one of: transponder power, transponder spectrum monitoring, transponder connectivity, transponder gain settings, transponder limiter settings, transponder automatic level control settings, transponder phase settings, internal gain generation, bandwidth for at least one beam, at least one frequency band for at least one of the at least one beam, transponder beamforming settings, effective isotropic radiation power (EIRP) for at least one of the at least one beam, transponder channels, or beam steering.

8. The method of claim 6, wherein the reconfiguring of the payload according to at least one of the unencrypted host commands or the unencrypted hosted commands comprises reconfiguring at least one of: at least one antenna, at least one analog-to-digital converter, at least one digital-to-analog converter, at least one beamformer, at least one digital channelizer, at least one demodulator, at least one modulator, at least one digital switch matrix, at least one digital combiner, or at least one

analog switch matrix.
9. The method of claim 6, wherein the vehicle is an airborne vehicle.
10. The method of claim 9, wherein the airborne vehicle is one of a satellite, aircraft, unmanned aerial vehicle (UAV), or space plane.
11. A method for a virtual transponder utilizing inband commanding, the method comprising: receiving, by a payload antenna on a vehicle via a host receiving antenna, encrypted hosted commands transmitted from a host spacecraft operations center (SOC), the encrypted hosted commands transmitted to the host SOC from a hosted payload (HoP) operation center (HOC), wherein the encrypted hosted commands are encrypted utilizing a second communication security (COMSEC) variety; receiving, by the vehicle, encrypted host commands transmitted from the host SOC, wherein the encrypted host commands are encrypted utilizing a first COMSEC variety; decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands; decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands; reconfiguring a payload on the vehicle according to at least one of the unencrypted host commands or the unencrypted hosted commands; transmitting, by the payload antenna, payload data to at least one of a host receiving antenna or the hosted receiving antenna; encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry; encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry; transmitting, by a host telemetry transmitter, the encrypted host telemetry to the host SOC; and transmitting, by a hosted telemetry transmitter via the host SOC, the encrypted hosted telemetry to the HOC.
12. The method of claim 11, wherein the reconfiguring of the payload according to at least one of the unencrypted host commands or the unencrypted hosted commands comprises adjusting at least one of: transponder power, transponder spectrum monitoring, transponder connectivity, transponder gain settings, transponder limiter settings, transponder automatic level control settings, transponder phase settings, internal gain generation, bandwidth for at least one beam, at least one frequency band for at least one of the at least one beam, transponder beamforming settings, effective isotropic radiation power (EIRP) for at least one of the at least one beam, transponder channels, or beam steering.
13. The method of claim 11, wherein the reconfiguring of the payload according to at least one of the unencrypted host commands or the unencrypted hosted commands comprises reconfiguring at least one of: at least one antenna, at least one analog-to-digital converter, at least one digital-to-analog converter, at least one beamformer, at least one digital channelizer, at least one demodulator, at least one modulator, at least one digital switch matrix, at least one digital combiner, or at least one analog switch matrix.
14. The method of claim 11, wherein the vehicle is an airborne vehicle.
15. The method of claim 14, wherein the airborne vehicle is one of a satellite, aircraft, unmanned aerial vehicle (UAV), or space plane.
16. A method for a virtual transponder utilizing inband commanding, the method comprising: receiving, by a payload antenna on a vehicle via a host receiving antenna, encrypted host commands and encrypted hosted commands transmitted from a host spacecraft operations center (SOC), the encrypted hosted commands transmitted to the host SOC from a hosted payload (HoP) operation center (HOC), wherein the encrypted host commands are encrypted utilizing a first communication security (COMSEC) variety, wherein the encrypted hosted commands are encrypted utilizing a second COMSEC variety; decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands; decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands;

reconfiguring a payload on the vehicle according to at least one of the unencrypted host commands or the unencrypted hosted commands; transmitting, by the payload antenna, payload data to at least one of the host receiving antenna or a hosted receiving antenna; encrypting, by the first communication security module, unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry; and transmitting, by a telemetry transmitter via the host SOC, the encrypted telemetry to the HOC.

17. The method of claim 16, wherein the reconfiguring of the payload according to at least one of the unencrypted host commands or the unencrypted hosted commands comprises adjusting at least one of: transponder power, transponder spectrum monitoring, transponder connectivity, transponder gain settings, transponder limiter settings, transponder automatic level control settings, transponder phase settings, internal gain generation, bandwidth for at least one beam, at least one frequency band for at least one of the at least one beam, transponder beamforming settings, effective isotropic radiation power (EIRP) for at least one of the at least one beam, transponder channels, or beam steering.

18. The method of claim 16, wherein the reconfiguring of the payload according to at least one of the unencrypted host commands or the unencrypted hosted commands comprises reconfiguring at least one of: at least one antenna, at least one analog-to-digital converter, at least one digital-to-analog converter, at least one beamformer, at least one digital channelizer, at least one demodulator, at least one modulator, at least one digital switch matrix, at least one digital combiner, or at least one analog switch matrix.

19. A method for a virtual transponder utilizing inband commanding, the method comprising: receiving, by a payload antenna on a vehicle via a host receiving antenna, encrypted host commands and encrypted hosted commands transmitted from a host spacecraft operations center (SOC), the encrypted hosted commands transmitted to the host SOC from a hosted payload (HoP) operation center (HOC), wherein the encrypted host commands are encrypted utilizing a first communication security (COMSEC) variety, wherein the encrypted hosted commands are encrypted utilizing a second COMSEC variety; decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands; decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands; reconfiguring a payload on the vehicle according to at least one of the unencrypted host commands or the unencrypted hosted commands; transmitting, by the payload antenna, payload data to at least one of the host receiving antenna or a hosted receiving antenna; encrypting, by the first communication security module, unencrypted host telemetry utilizing the first COMSEC variety to generate encrypted host telemetry; encrypting, by the second communication security module, unencrypted hosted telemetry utilizing the second COMSEC variety to generate encrypted hosted telemetry; and transmitting, by the payload antenna via the host receiving antenna and the host SOC, the encrypted host telemetry and the encrypted hosted telemetry to the HOC.

20. A method for a virtual transponder utilizing inband commanding, the method comprising: receiving, by a payload antenna on a vehicle via a host receiving antenna, encrypted host commands and encrypted hosted commands transmitted from a host spacecraft operations center (SOC), the encrypted hosted commands transmitted to the host SOC from a hosted payload (HoP) operation center (HOC), wherein the encrypted host commands are encrypted utilizing a first communication security (COMSEC) variety, wherein the encrypted hosted commands are encrypted utilizing a second COMSEC variety; decrypting, by a first communication security module, the encrypted host commands utilizing the first COMSEC variety to generate unencrypted host commands; decrypting, by a second communication security module, the encrypted hosted commands utilizing the second COMSEC variety to generate unencrypted hosted commands; reconfiguring a payload on the vehicle according to at least one of the unencrypted host commands or the unencrypted hosted commands; transmitting, by the payload antenna, payload data to at least

one of the host receiving antenna or a hosted receiving antenna; encrypting, by the first communication security module, unencrypted telemetry utilizing the first COMSEC variety to generate encrypted telemetry; and transmitting, by the payload antenna via the host receiving antenna and the host SOC, the encrypted telemetry to the HOC.